PROGRAMA DE PÓS-GRADUAÇÃO EM

Biologia Vegetal

Universidade Federal de Pernambuco

O DNA e a cromatina em 3D

Amanda Souza Câmara

Leibniz Institute of Plant Genetics and Crop Plant Research



26 de outubro de 2026

In Silico Genebank Proteomics

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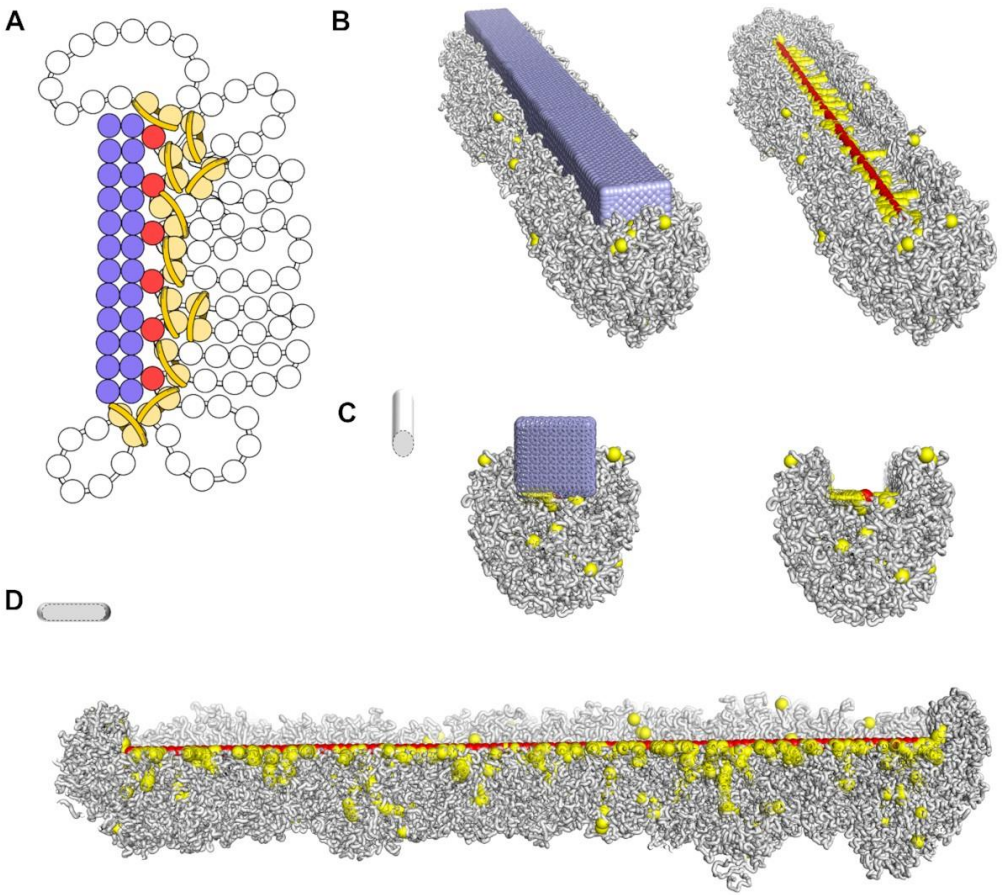


Eu no mundo

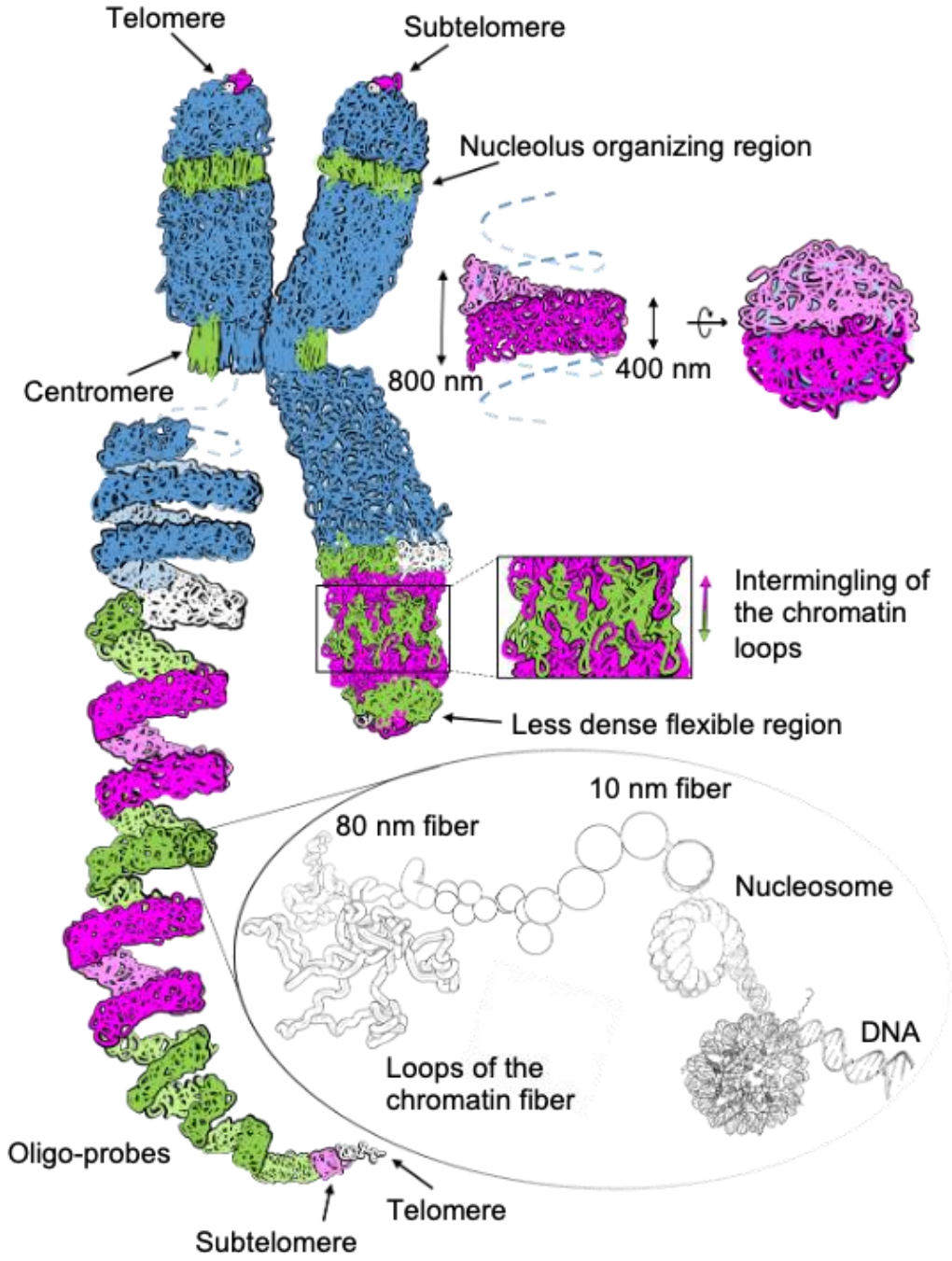
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UNIVERSIDADE DE SÃO PAULO
Instituto de Física de São Carlos

Meus highlights

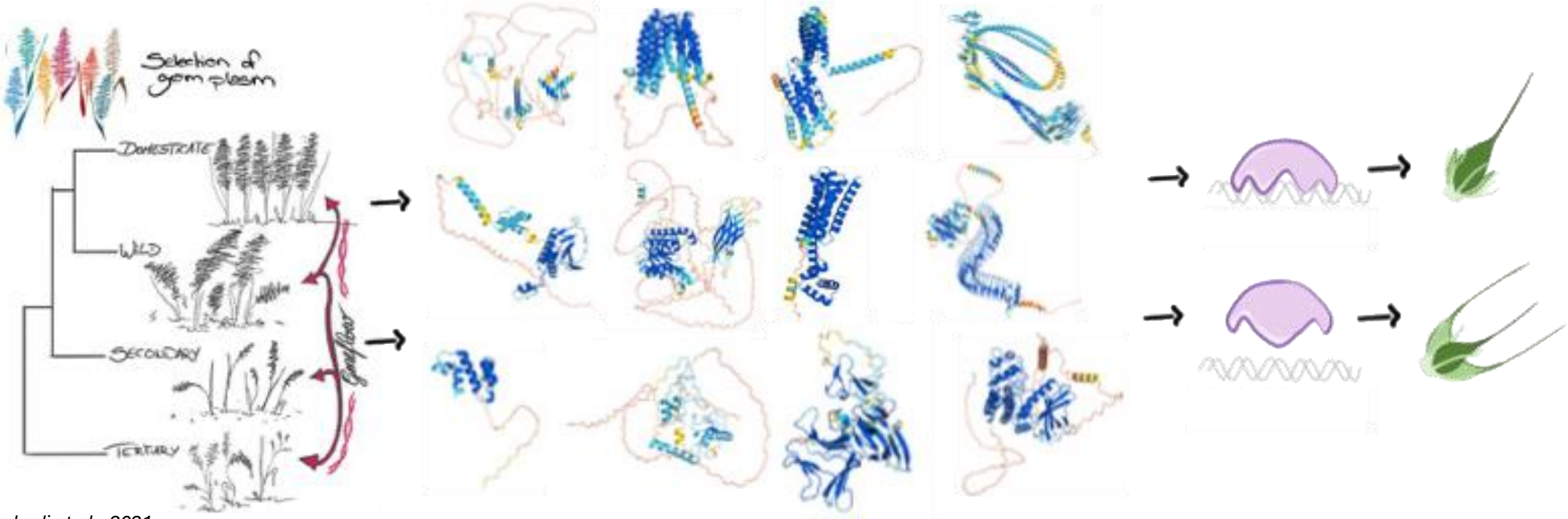


Câmara et al. 2021



Kubalová et al. 2023

Meus highlights



Jayakodi et al., 2021

Vocês?

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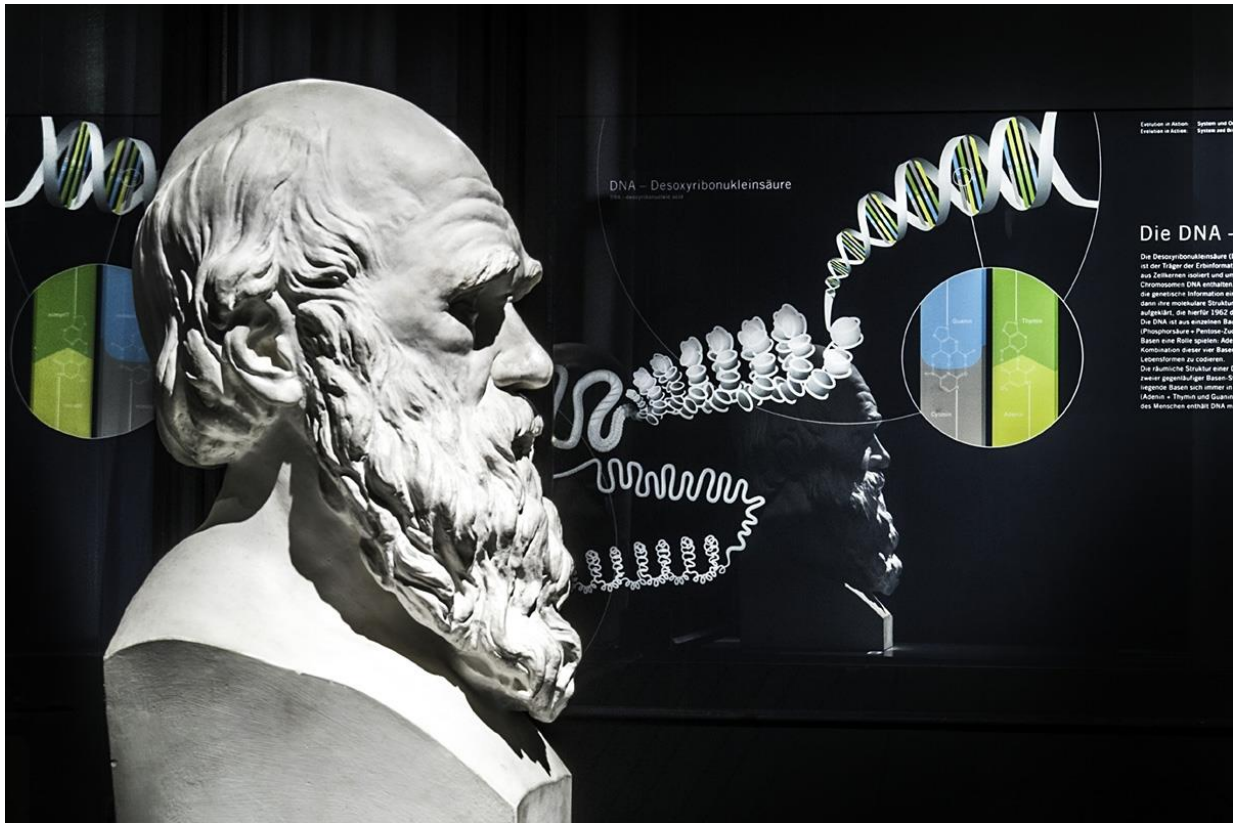
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Como o genoma se organiza?



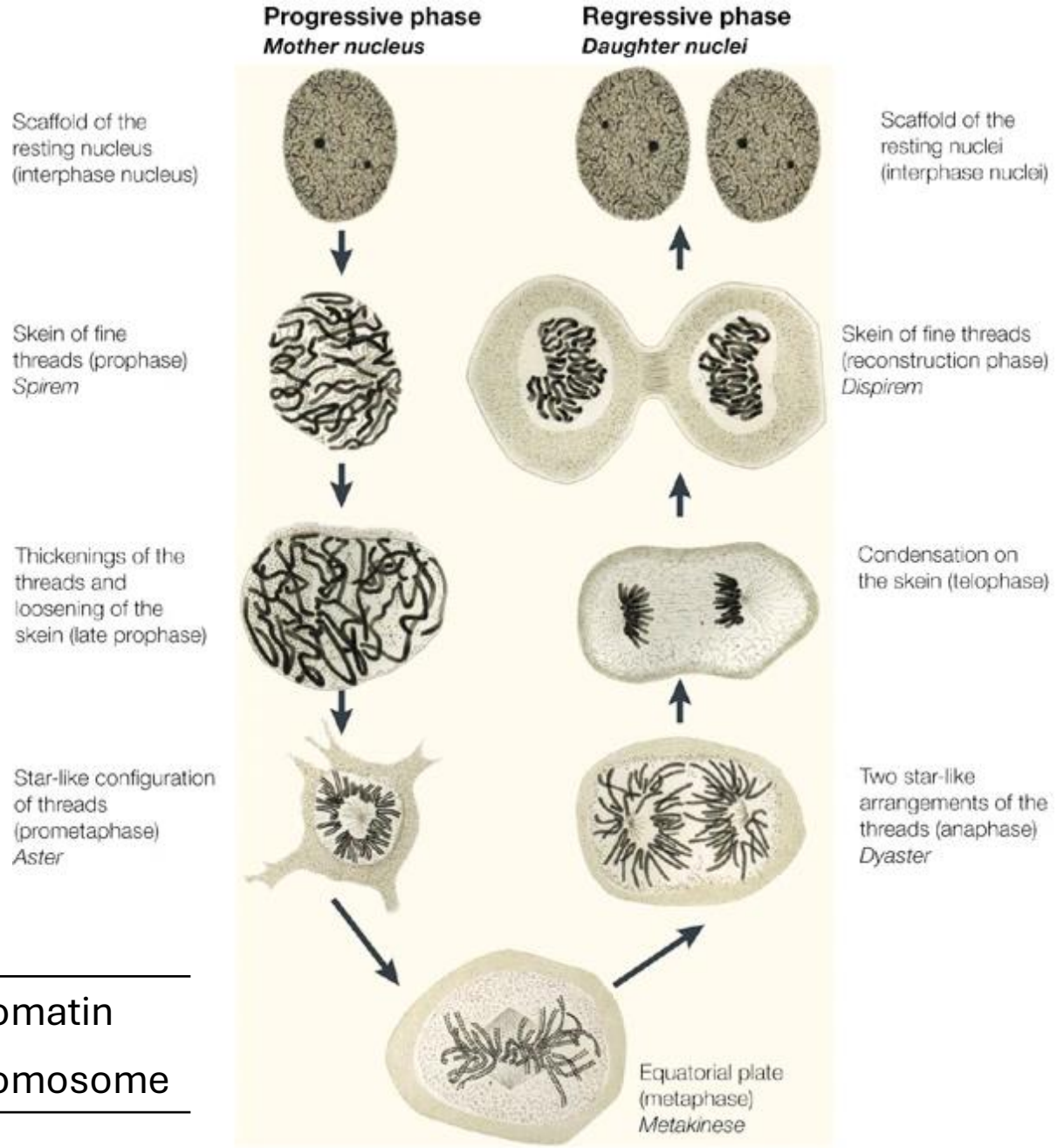
Iyer, 2011



Museu de história natural, Berlim

Genoma humano: 2m
Diâmetro do núcleo: 10µm

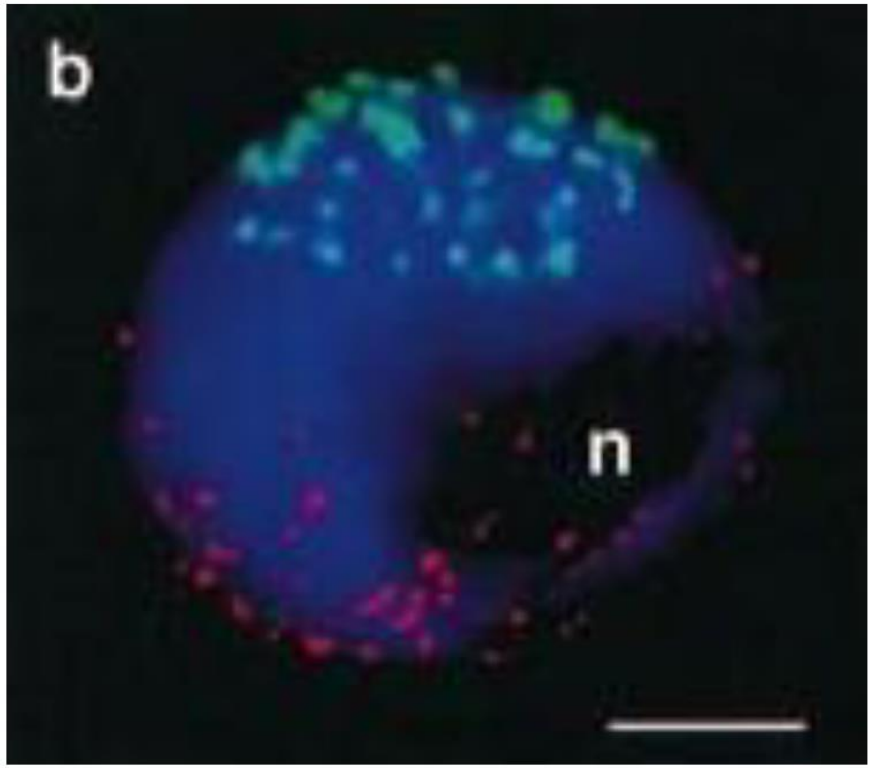
Coisas cromáticas

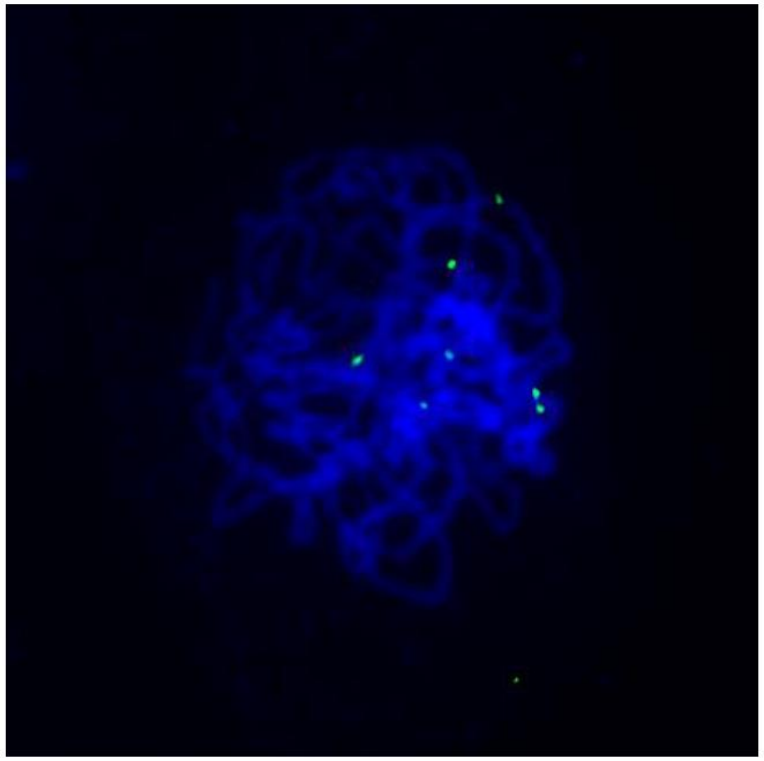


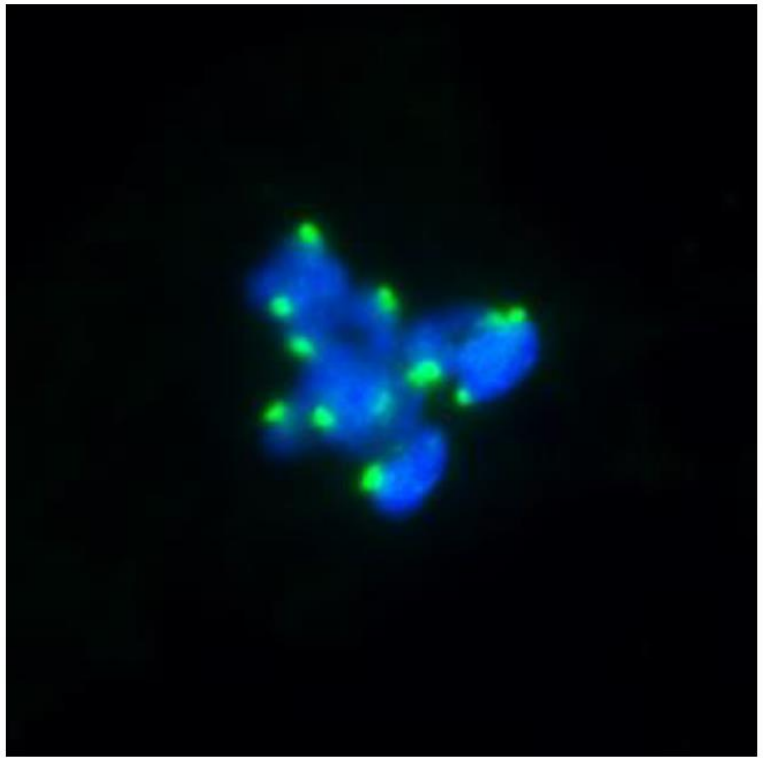
Coining “chromo” terms

~1880	Walther Flemming	Chromatin
1888	Heinrich Wilhelm Waldeyer	Chromosome

Como DAPI se liga ao DNA e por que preferencialmente a adenina e timina?

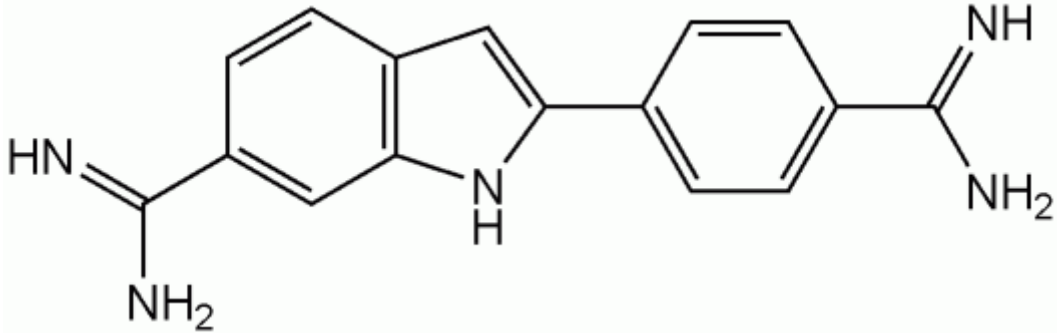




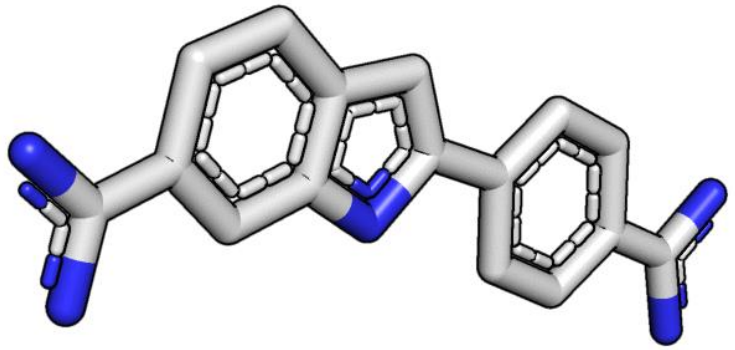


Dong & Jian, 1998.

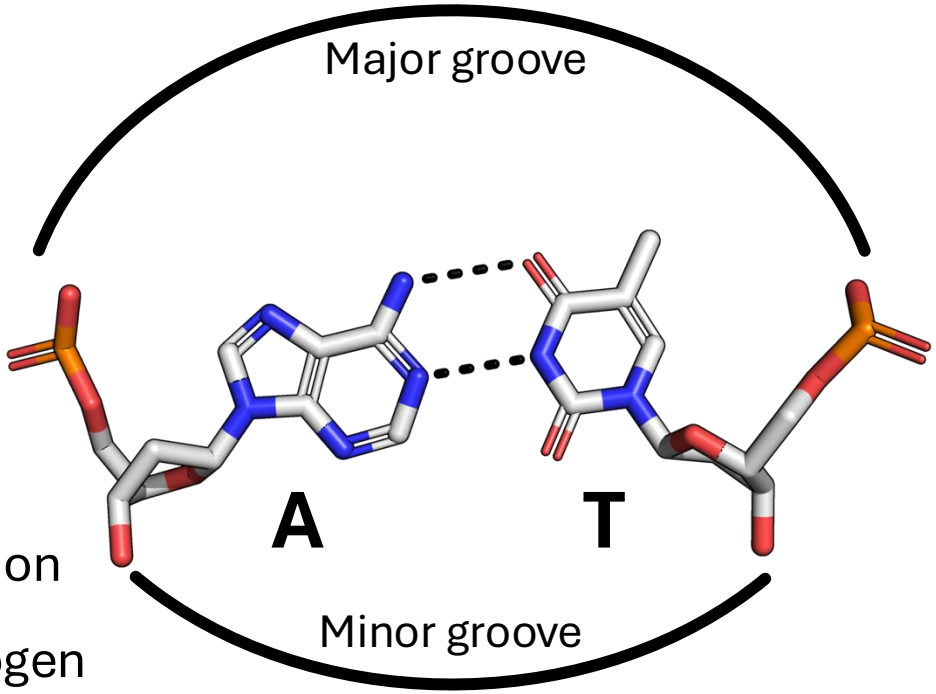
Structure of 4',6'-diamino-2-fenil-indol, DAPI

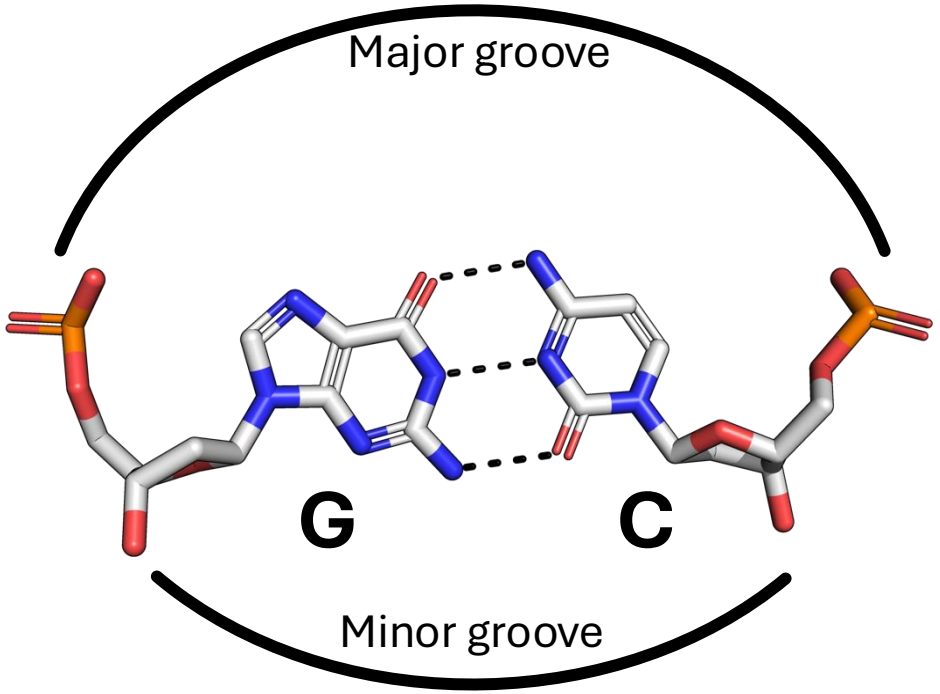


- Carbon
- Nitrogen
- Oxigen
- Phosphor



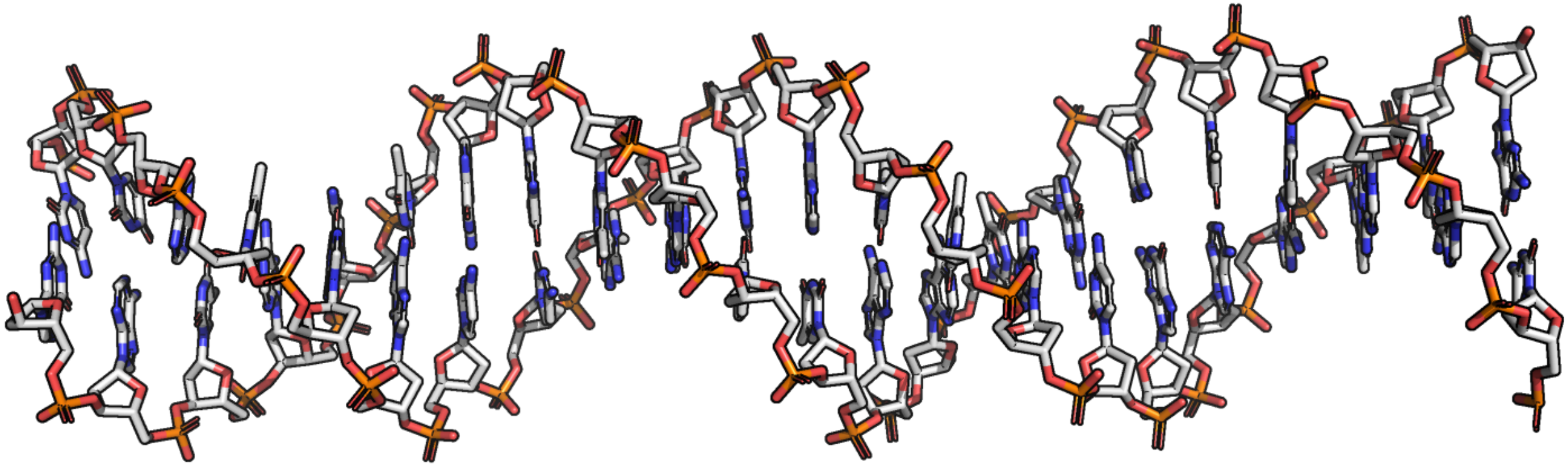
Structure of DNA





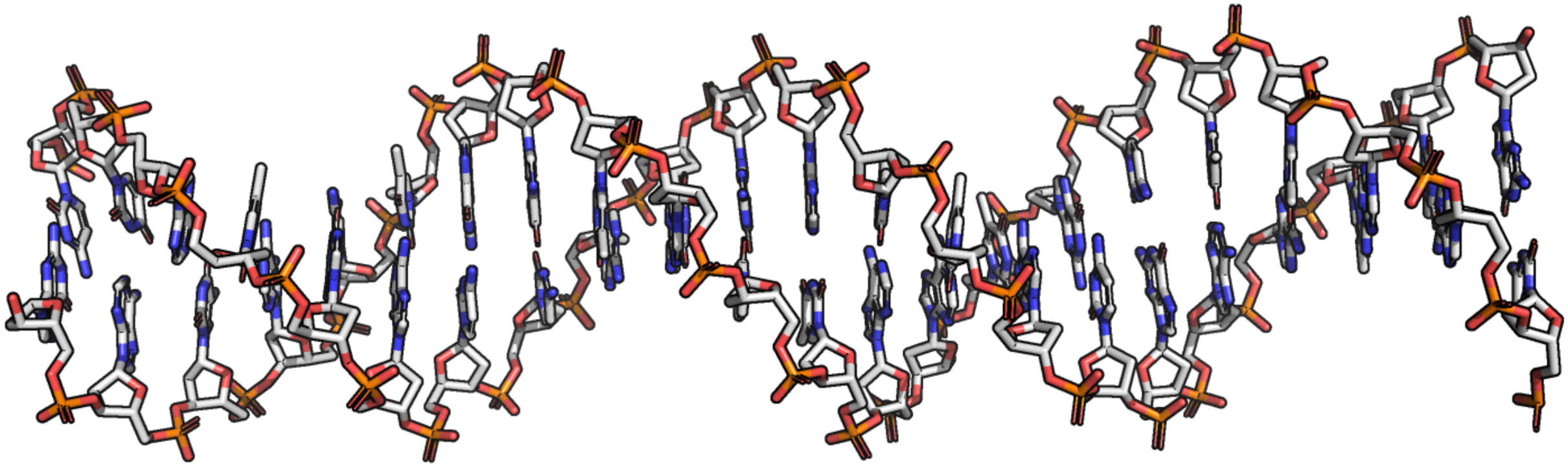
- Carbon
- Nitrogen
- Oxigen
- Phosphor

Structure of DNA



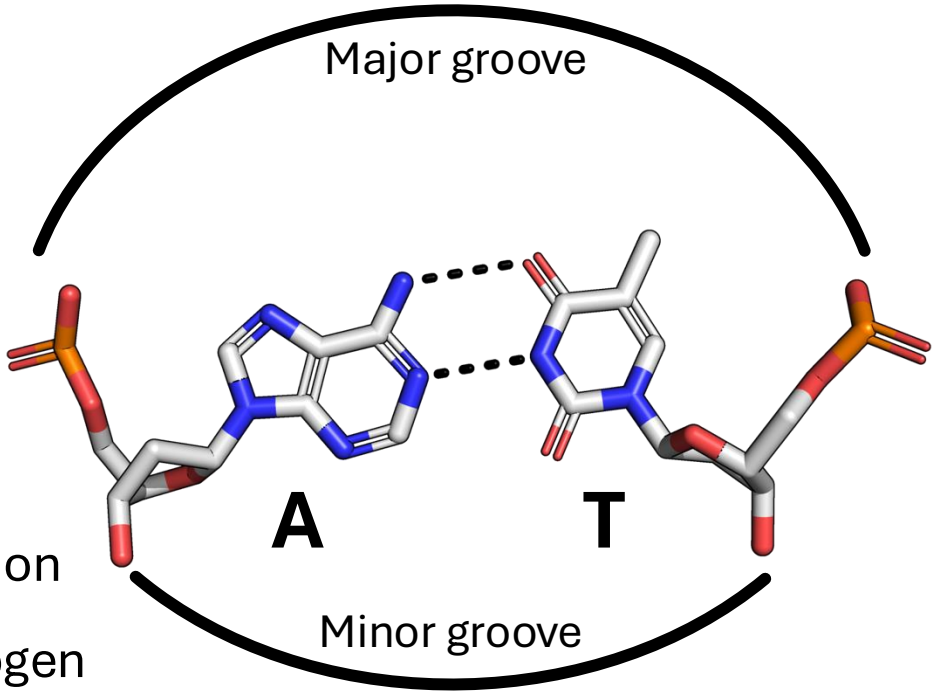
- Carbon
- Nitrogen
- Oxigen
- Phosphor

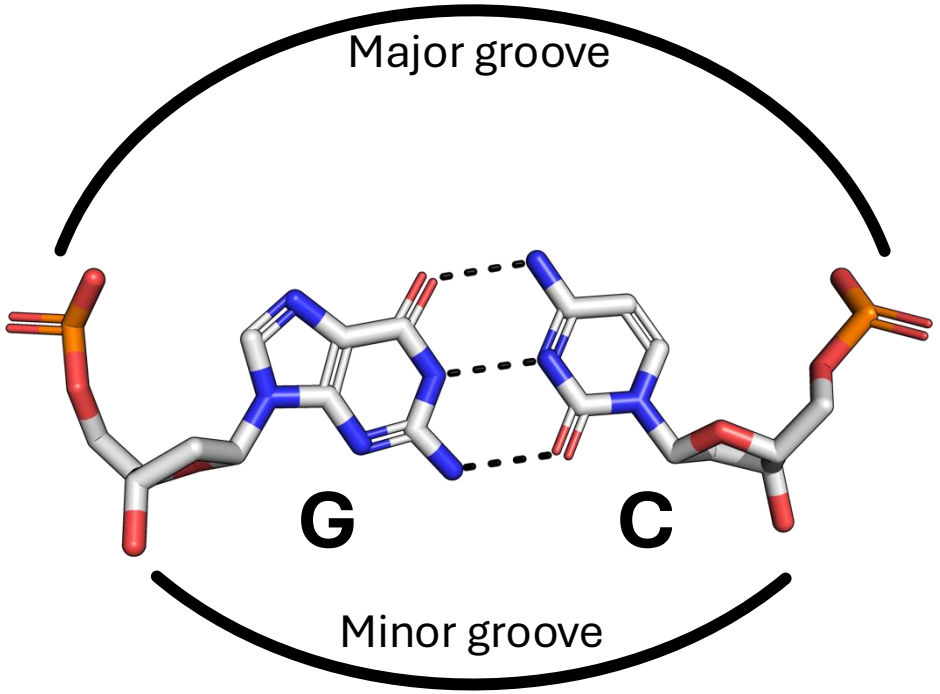
Structure of DNA



- Carbon
- Nitrogen
- Oxigen
- Phosphor

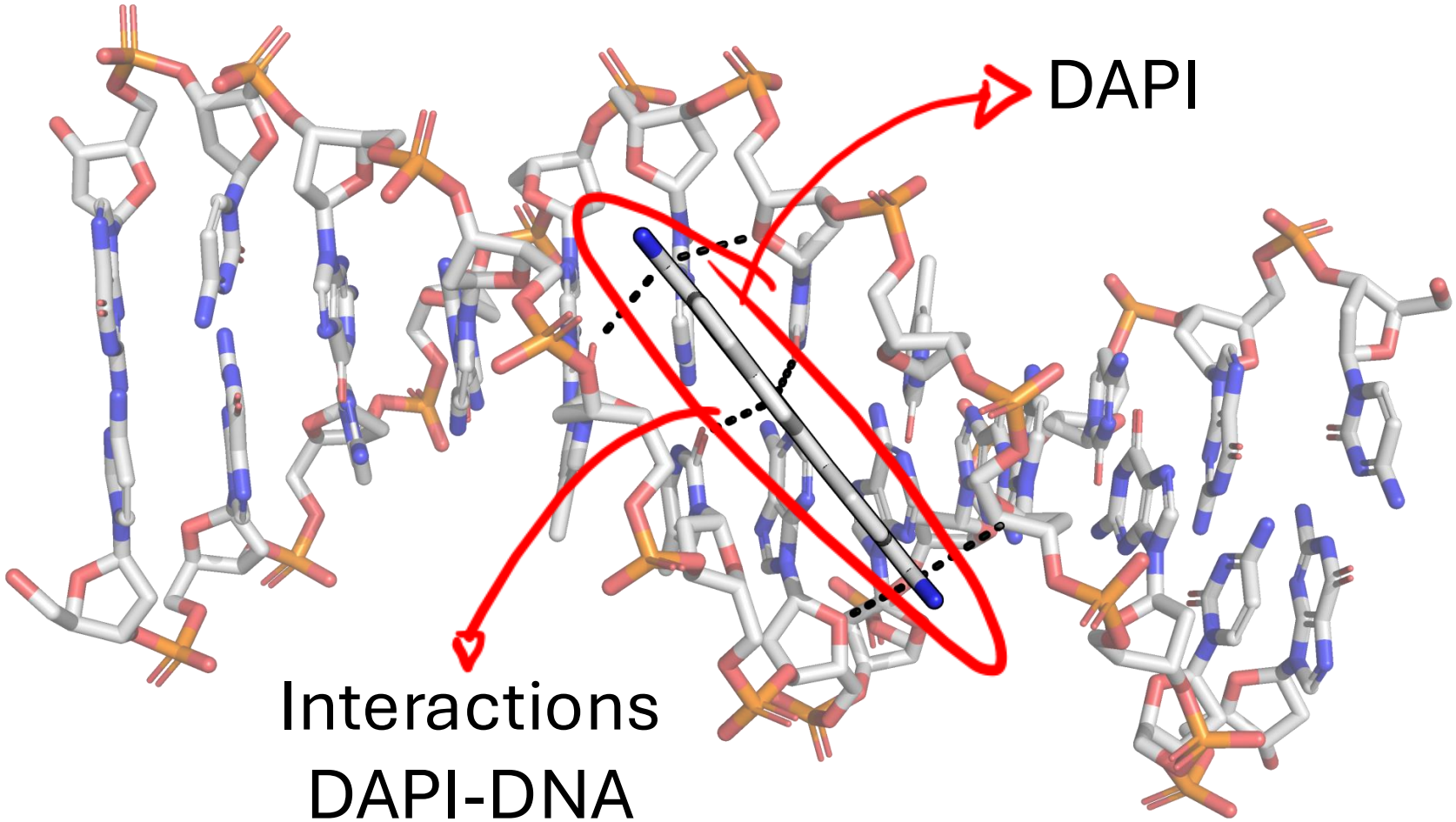
Structure of DNA





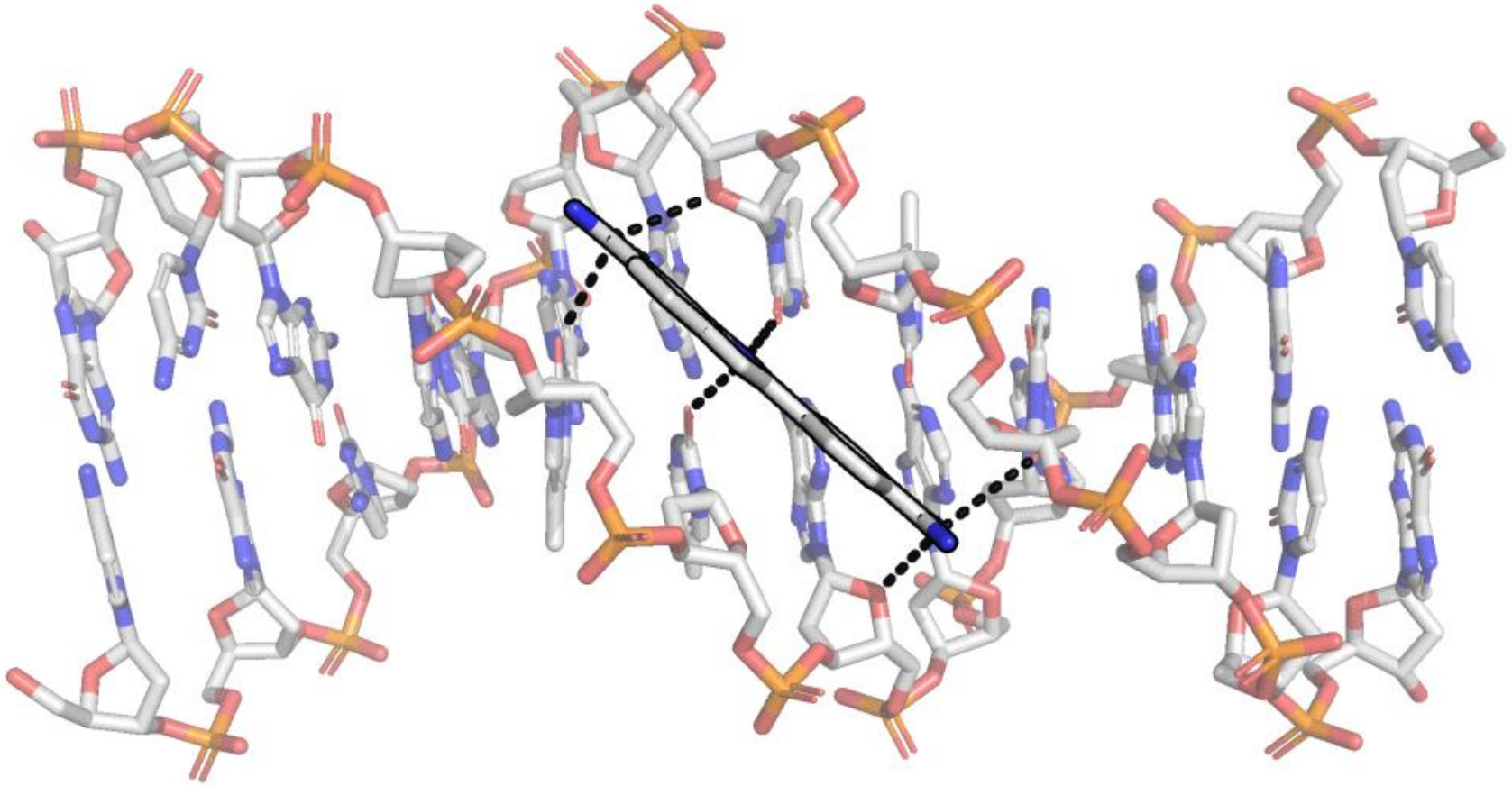
- Carbon
- Nitrogen
- Oxigen
- Phosphor

Structure of the DAPI-DNA complex



Ver no PyMOL PDB 1D30, 1989

Structure of the DAPI-DNA complex



PDB 1D30, 1989

O banco de dados de proteína



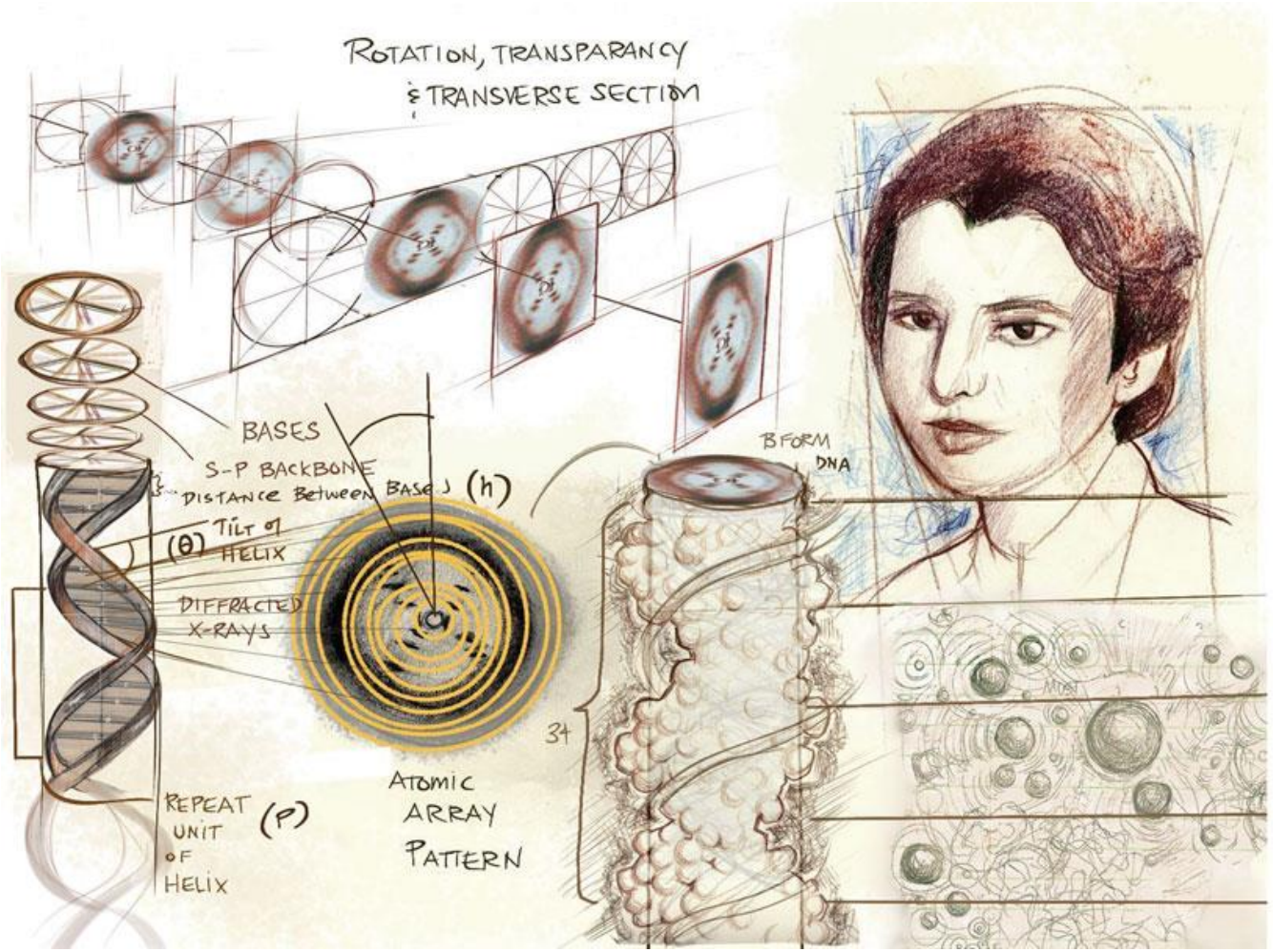
https://www.rcsb.org/
https://www.ebi.ac.uk/pdbe/
https://pdbj.org

Outros corantes

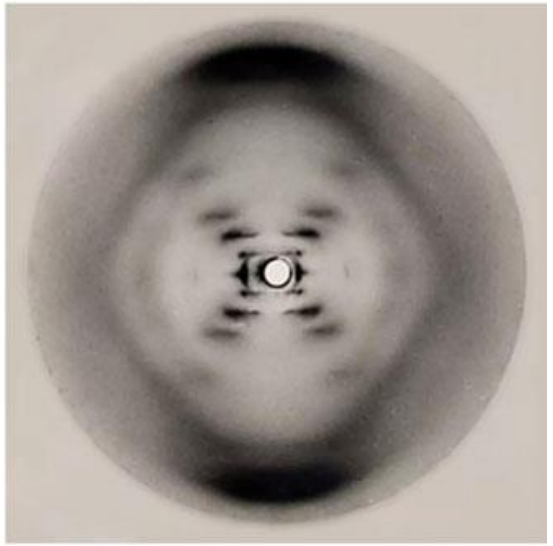
*Feito com IA

Staining molecule	Binding preference	PDB ID
DAPI	AT-rich DNA; minor groove	1D30, 432D
Hoechst 33258	AT-rich DNA; minor groove	1DNH
7-AAD (7-aminoactinomycin D)	GC-sensitive intercalating dye; binding depends on sequence context	1UNJ, 1UNM (non-complementary DNA)
Chromomycin A3	GC-rich DNA; minor groove, with divalent metal ions	1D83, 1EKH, 1EKI, 5YZE
Mithramycin A	GC-rich DNA; minor groove, with metal-ion coordination	146D

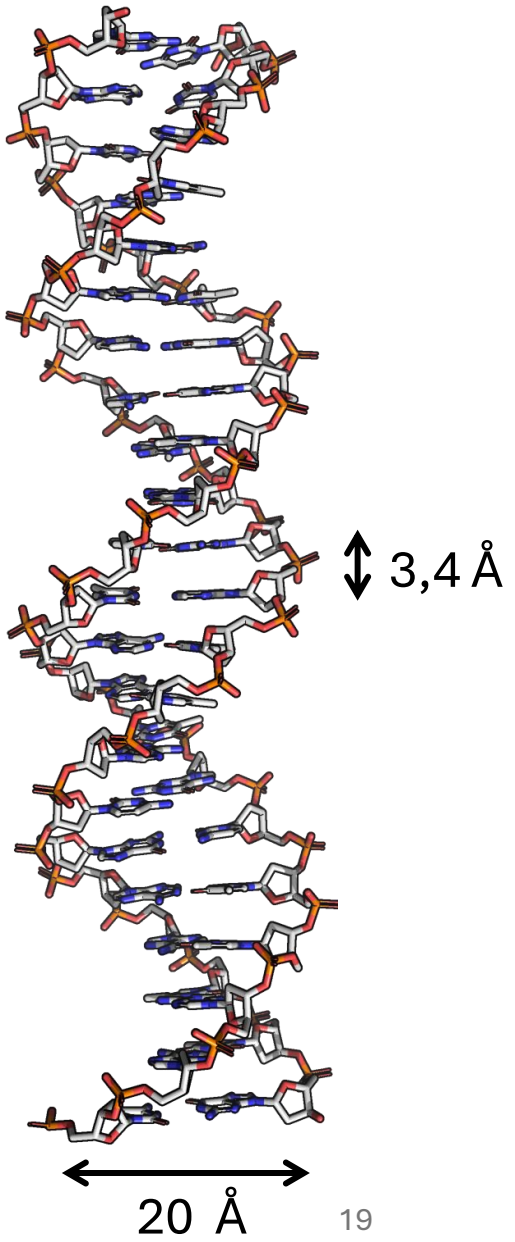
Solved by crystallography



Babaian, 2024



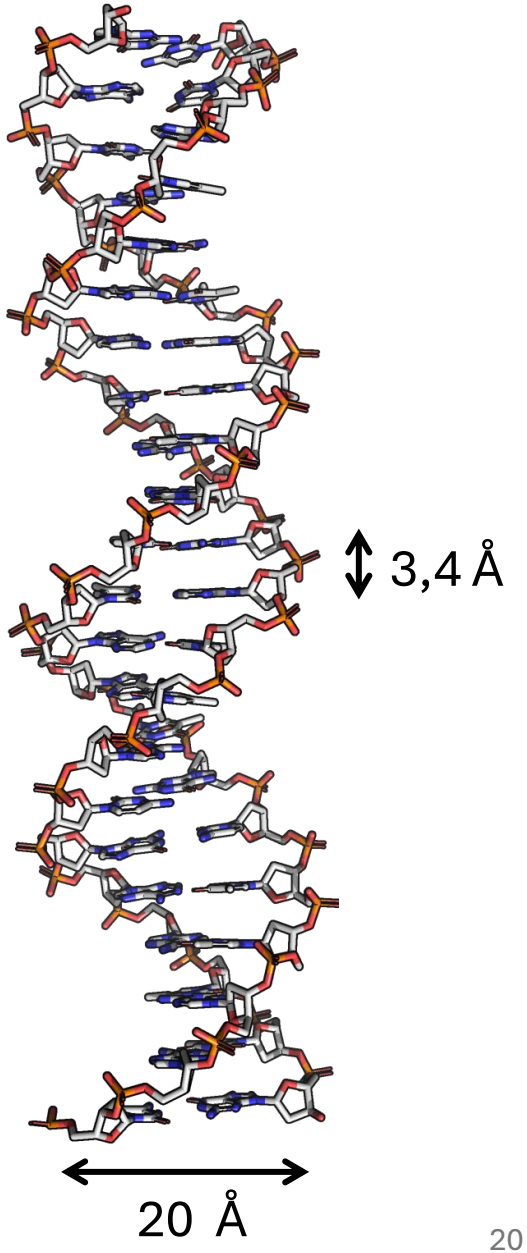
R. E. Franklin and R. Gosling, 1953



Um polímero muito muito longo

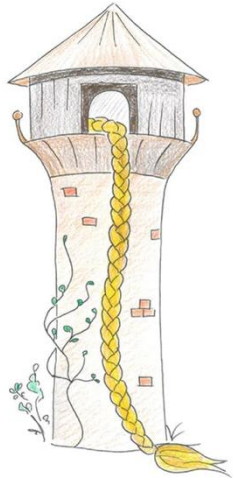
Objeto	Espessura	Comprimento
1 Mbp DNA	20 Å	0,34 mm

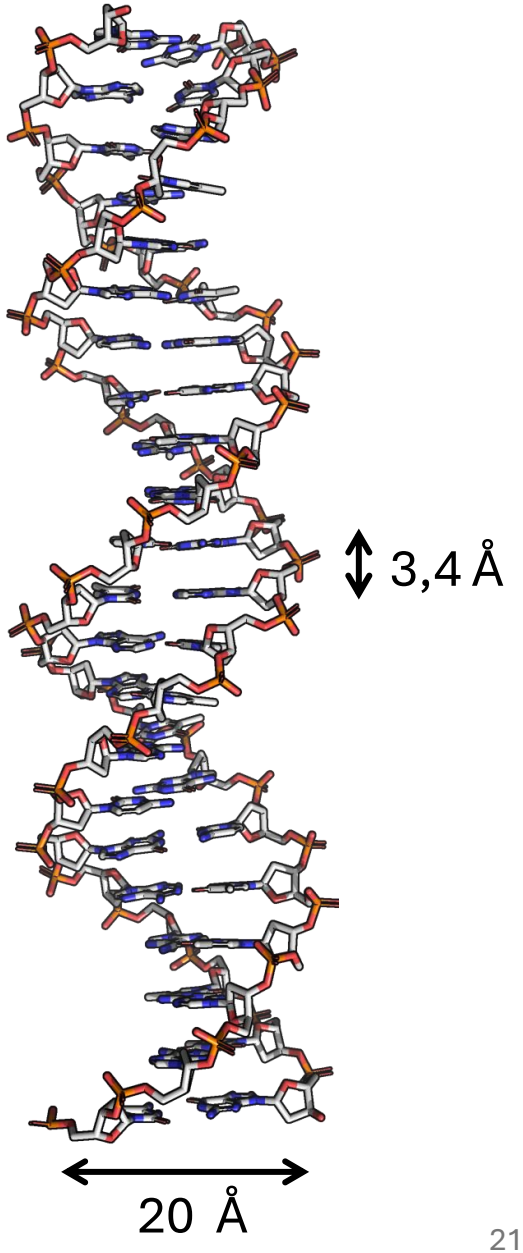
170 000 mais longo
que seu diâmetro!



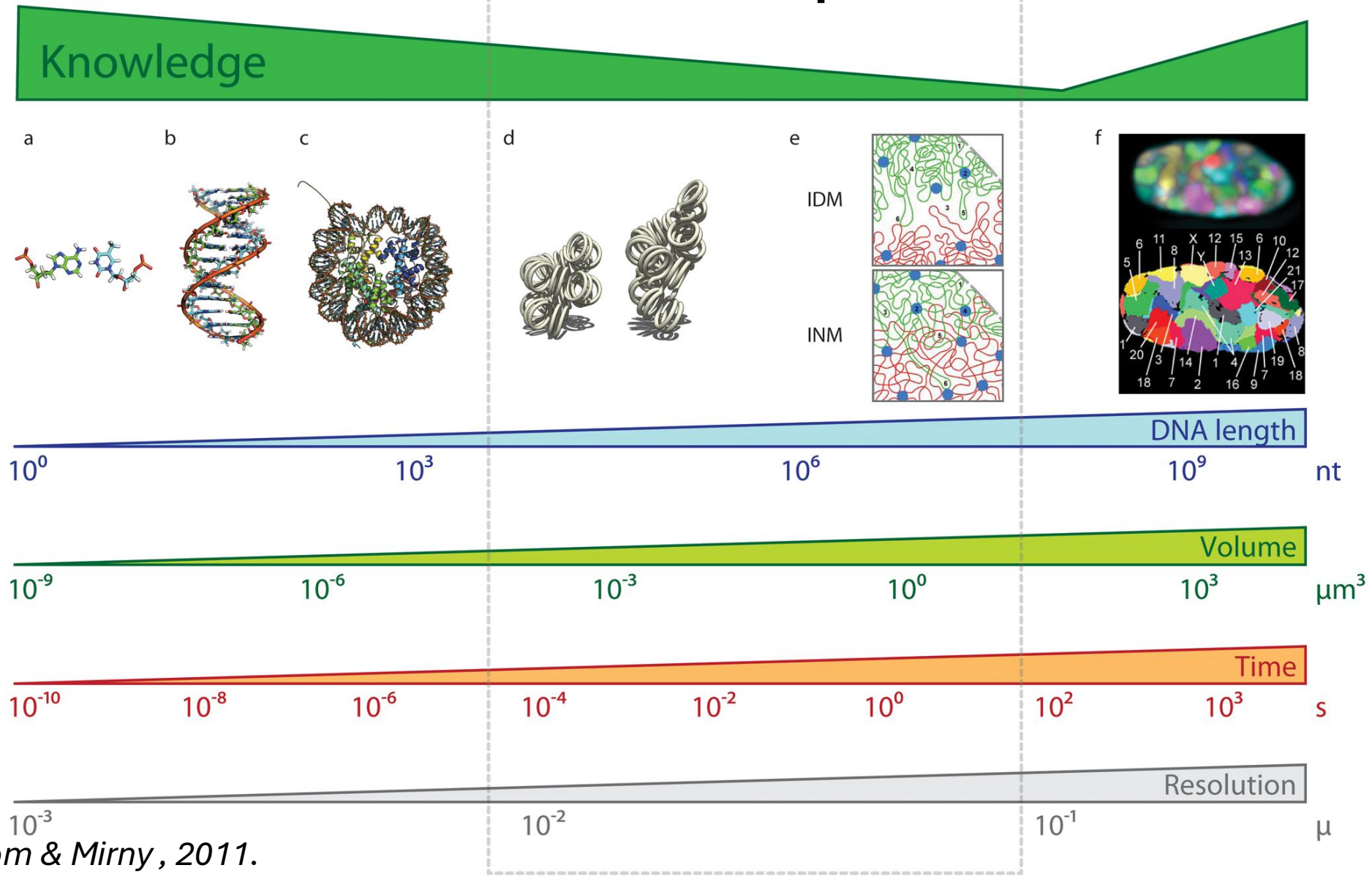
Um polímero muito muito longo

Objeto	Espessura	Comprimento
1 Mbp DNA	20 Å	0,34 mm
Um fio de cabelo	0,1 mm	17 m



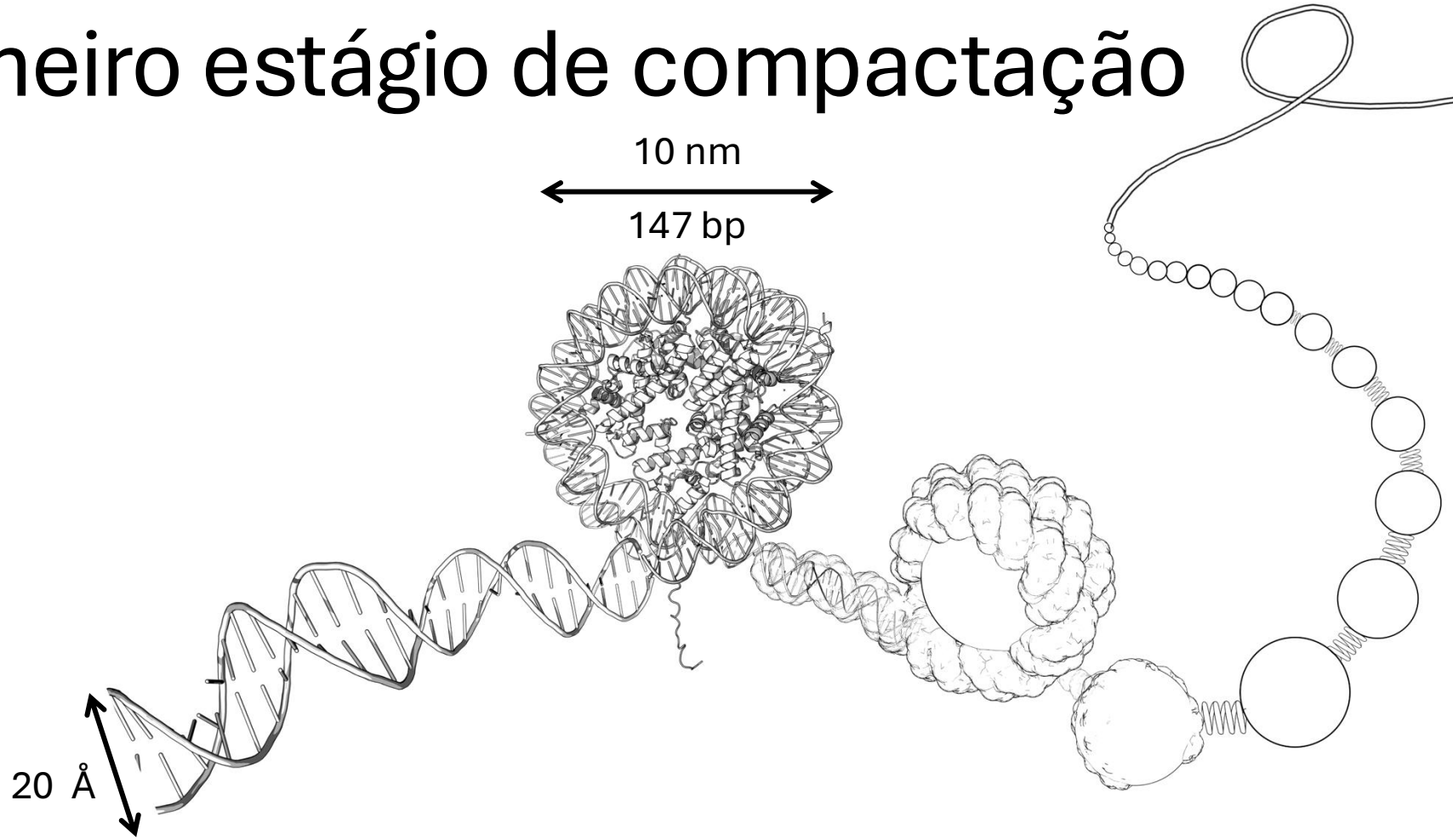


O DNA cobre escalas amplas



Marti-Renom & Mirny , 2011.

Primeiro estágio de compactação

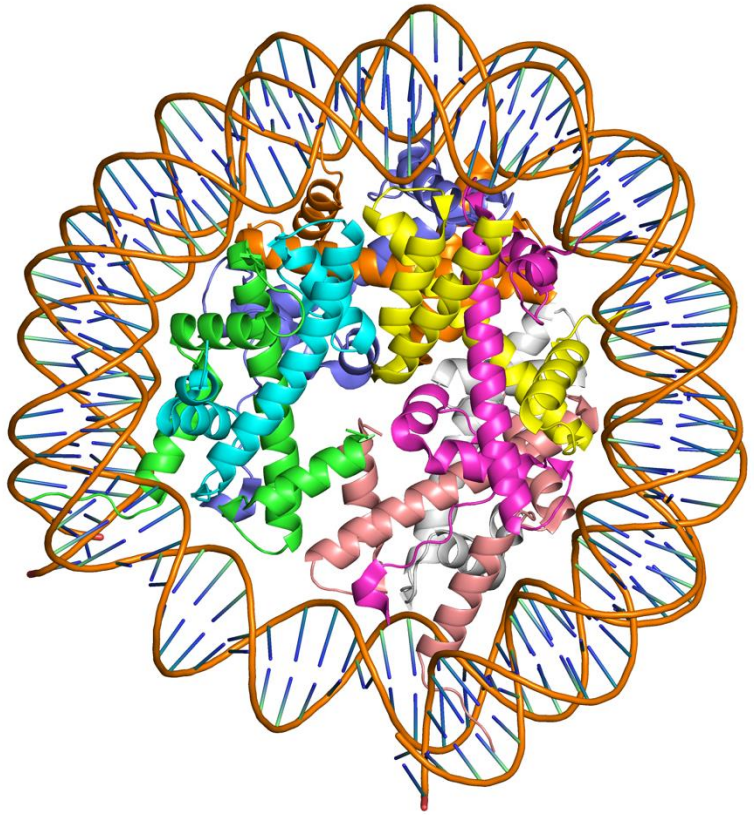


A fibra de cromatina também é um polímero muito muito longo.

Objeto	Espessura	Comprimento
1 Mbp DNA	20 Å	0,34 mm
1 Mbp cromatina	10 nm	50 µm

5 000 vezes mais longo que seu diâmetro!

O nucleossomo



Ver no PyMOL PDB 9k3z, 2024

AlphaFold vai mudar tudo

nature

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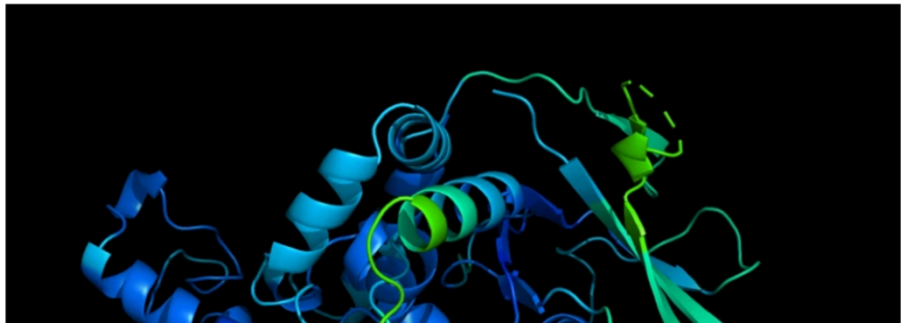
nature > news > article

NEWS | 30 November 2020

'It will change everything': DeepMind's AI makes gigantic leap in solving protein structures

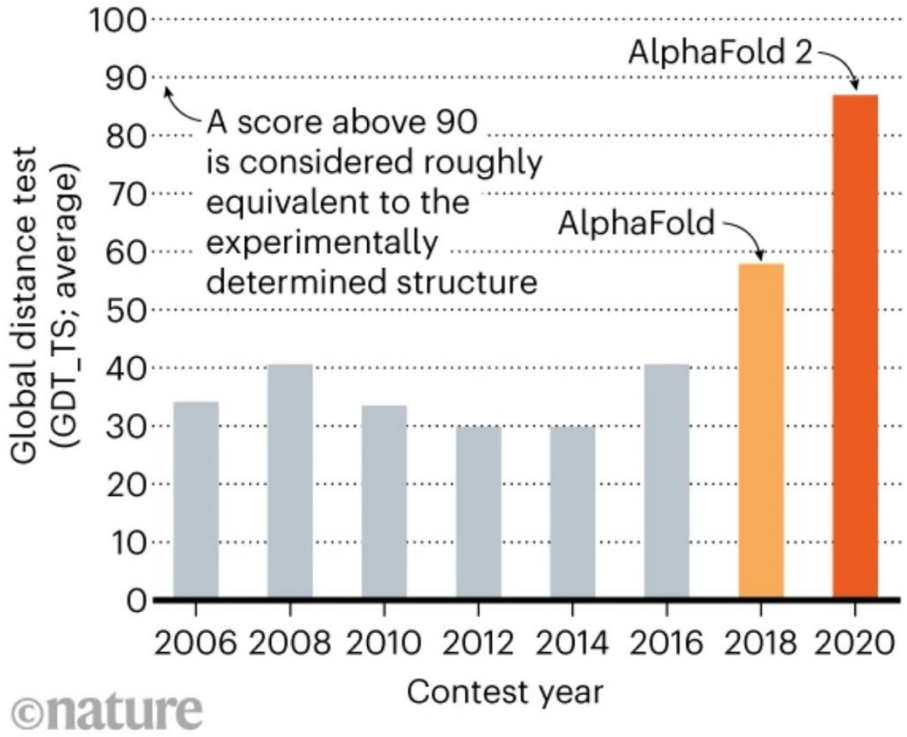
Google's deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.

Ewen Callaway

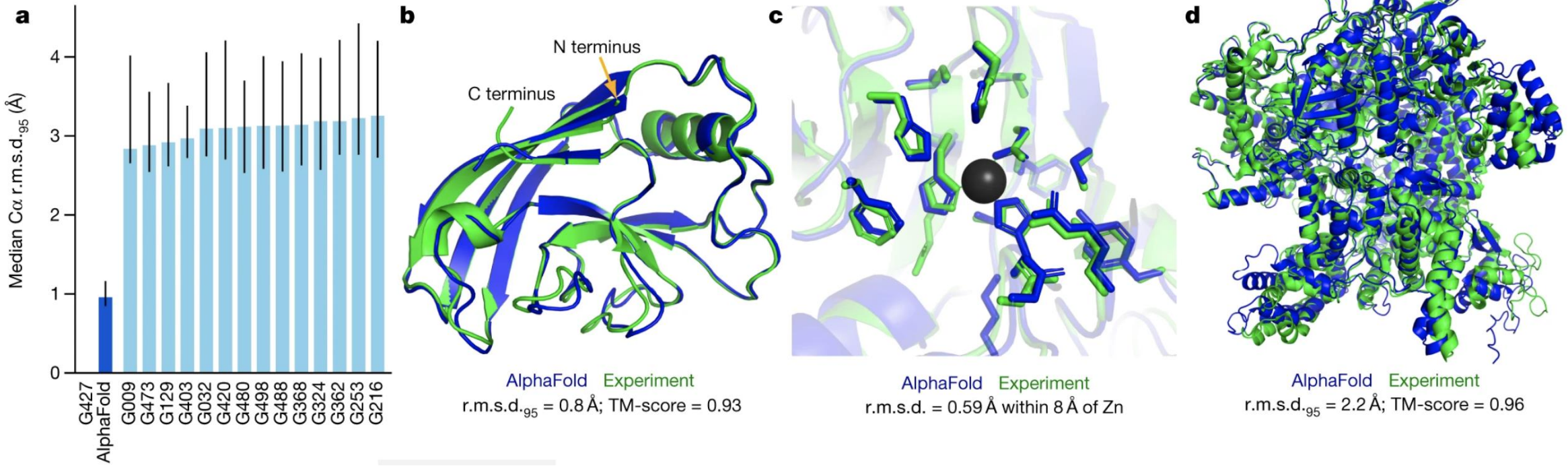


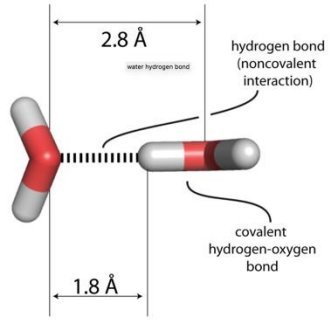
STRUCTURE SOLVER

DeepMind's AlphaFold 2 algorithm significantly outperformed other teams at the CASP14 protein-folding contest — and its previous version's performance at the last CASP.



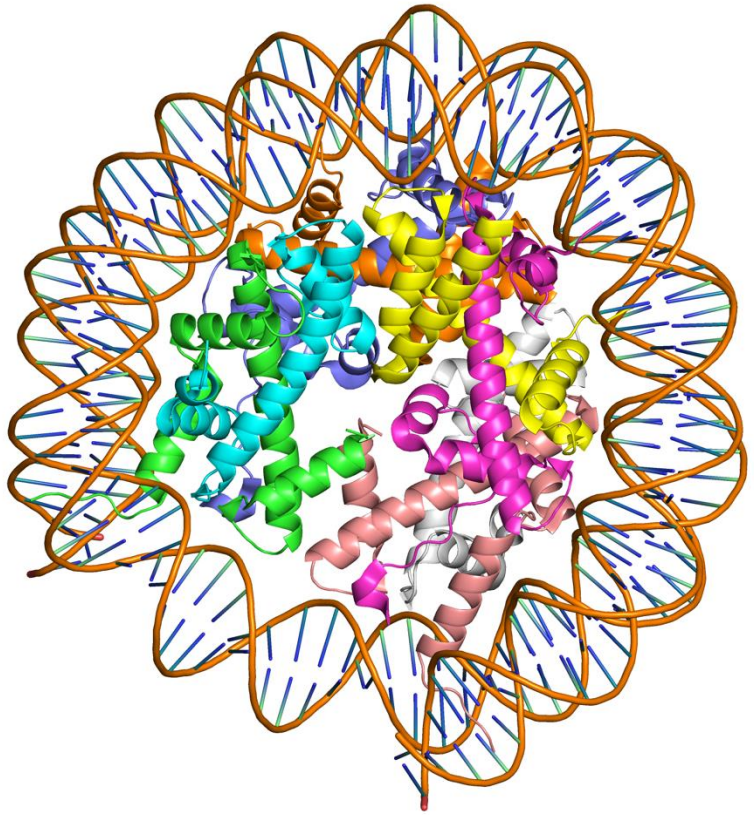
Precisão do AlphaFold

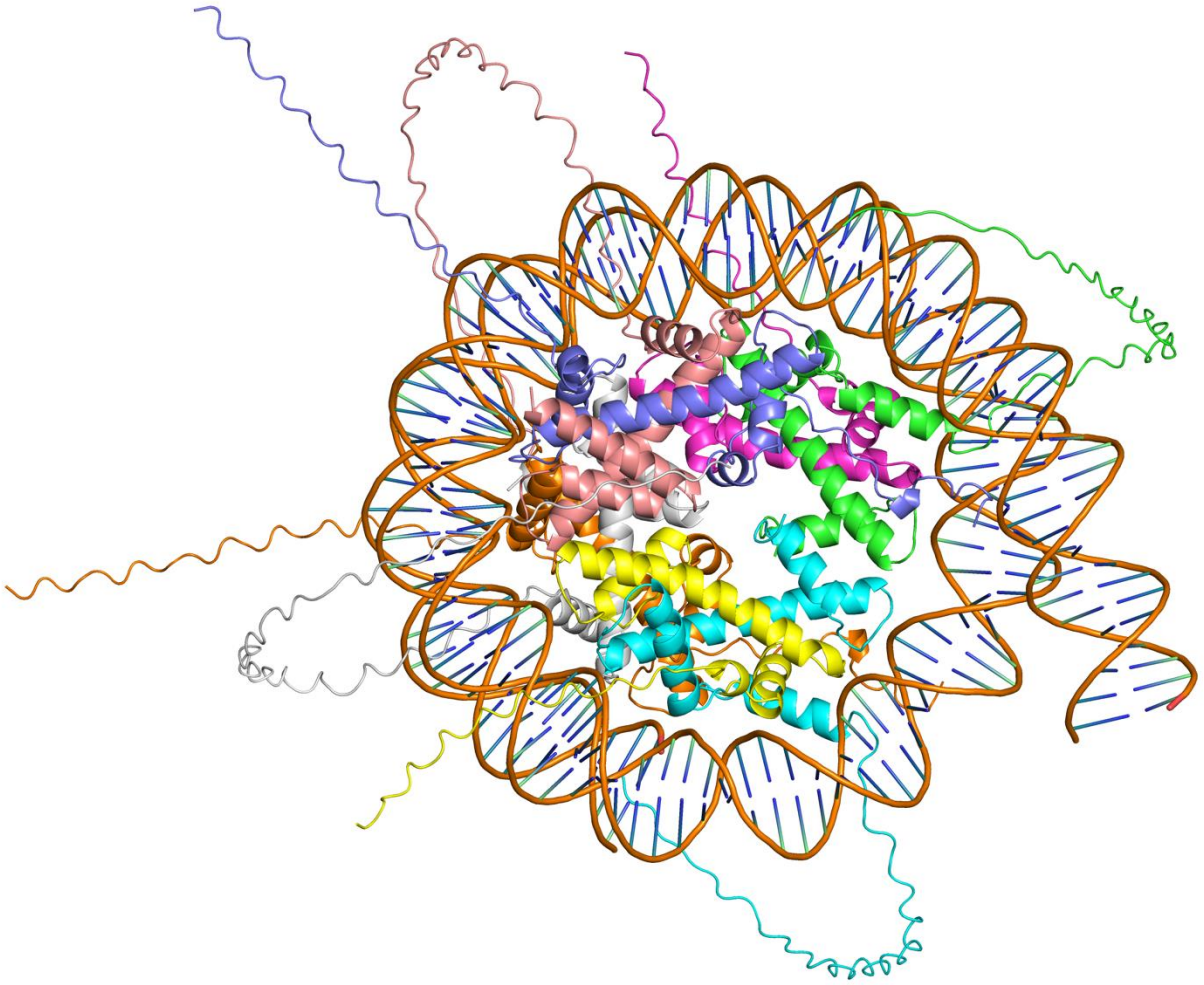




Jumper et al., 2021

O nucleossomo

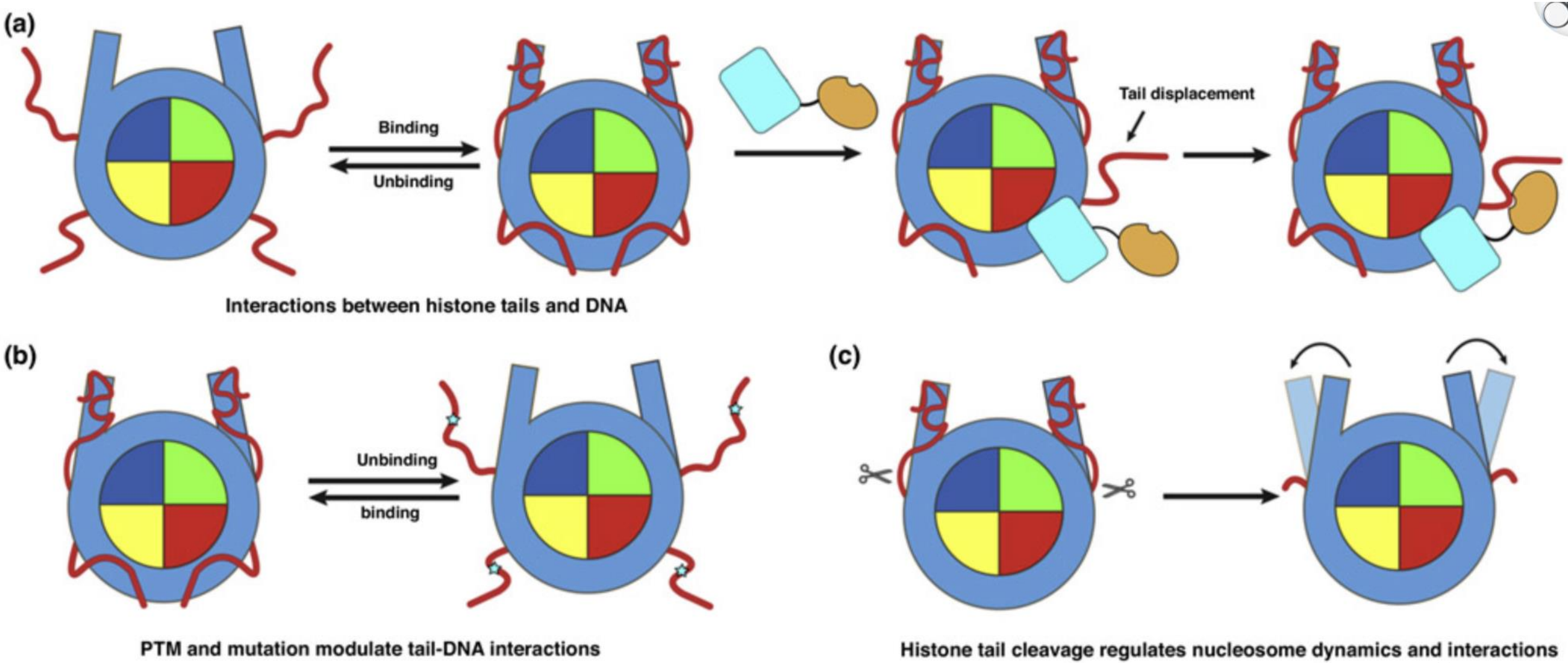




Ver no PyMOL PDB 9k3z, 2024

https://github.com/InsilicoGenebankProteomics/CursoOrganizacaoDoGenomaUFPE2026

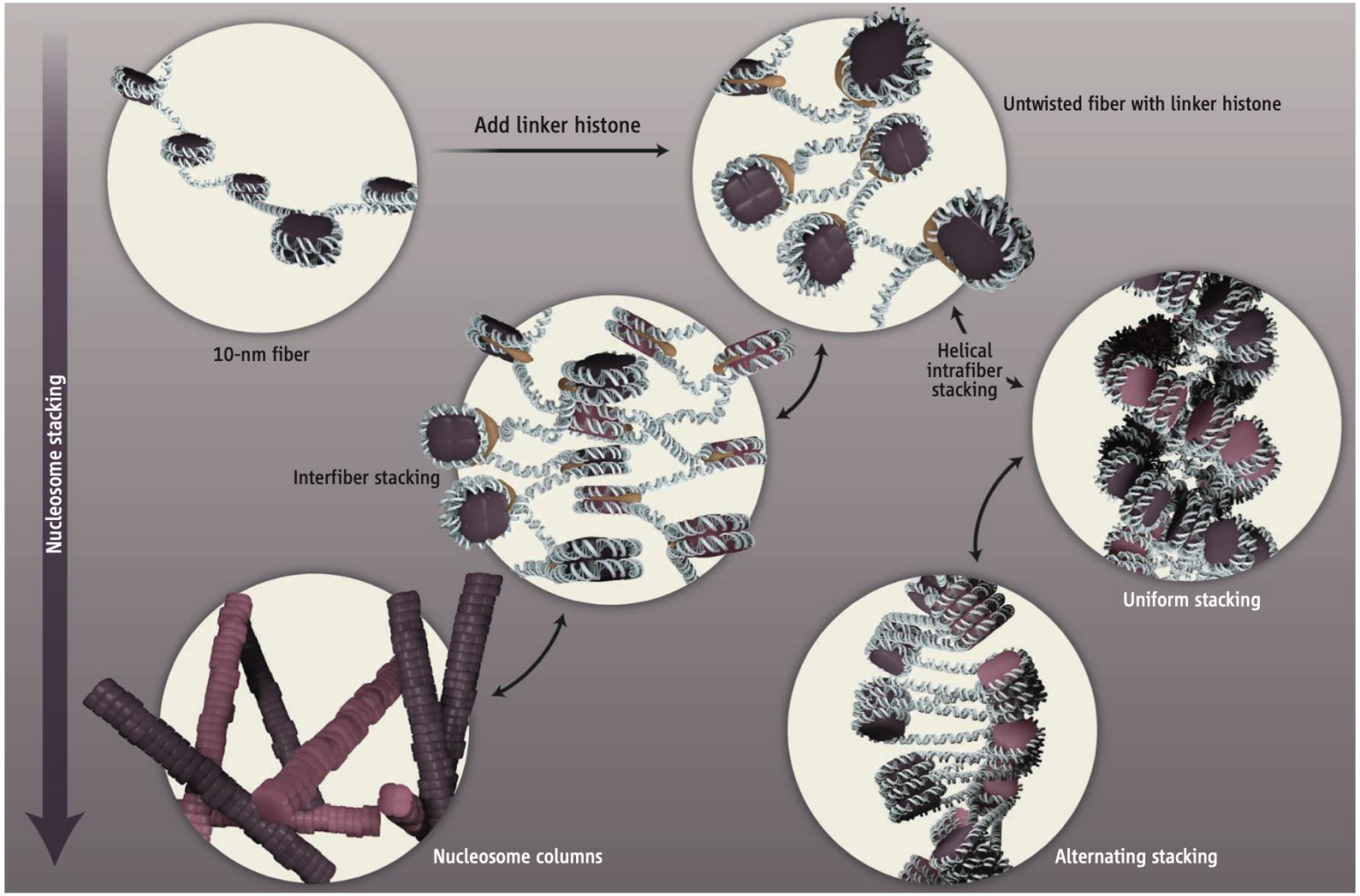
Papel das caudas de histonas



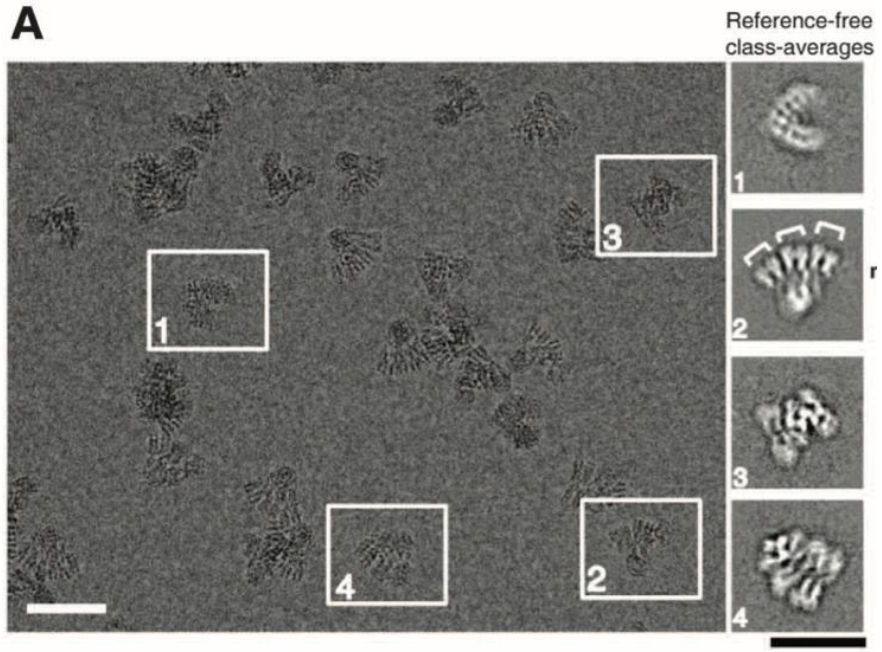
Peng et al. 2021

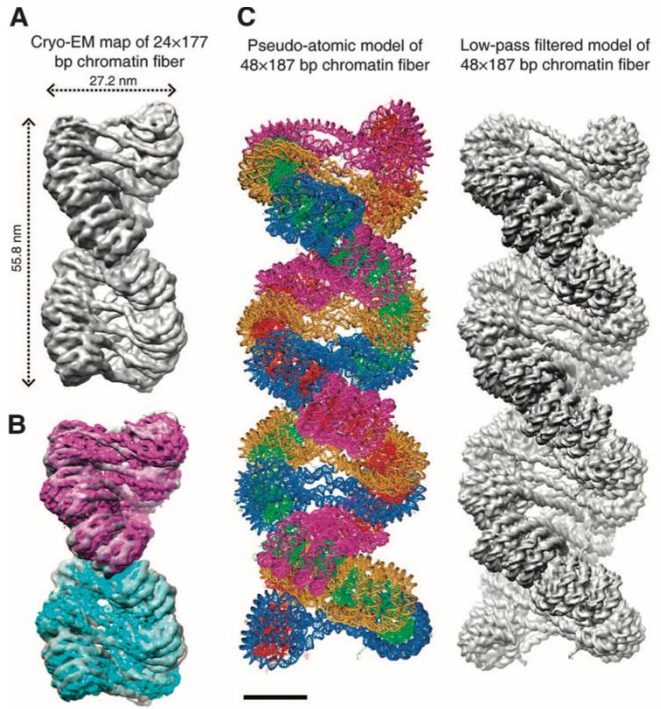
Current Opinion in Structural Biology

A fibra de cromatina



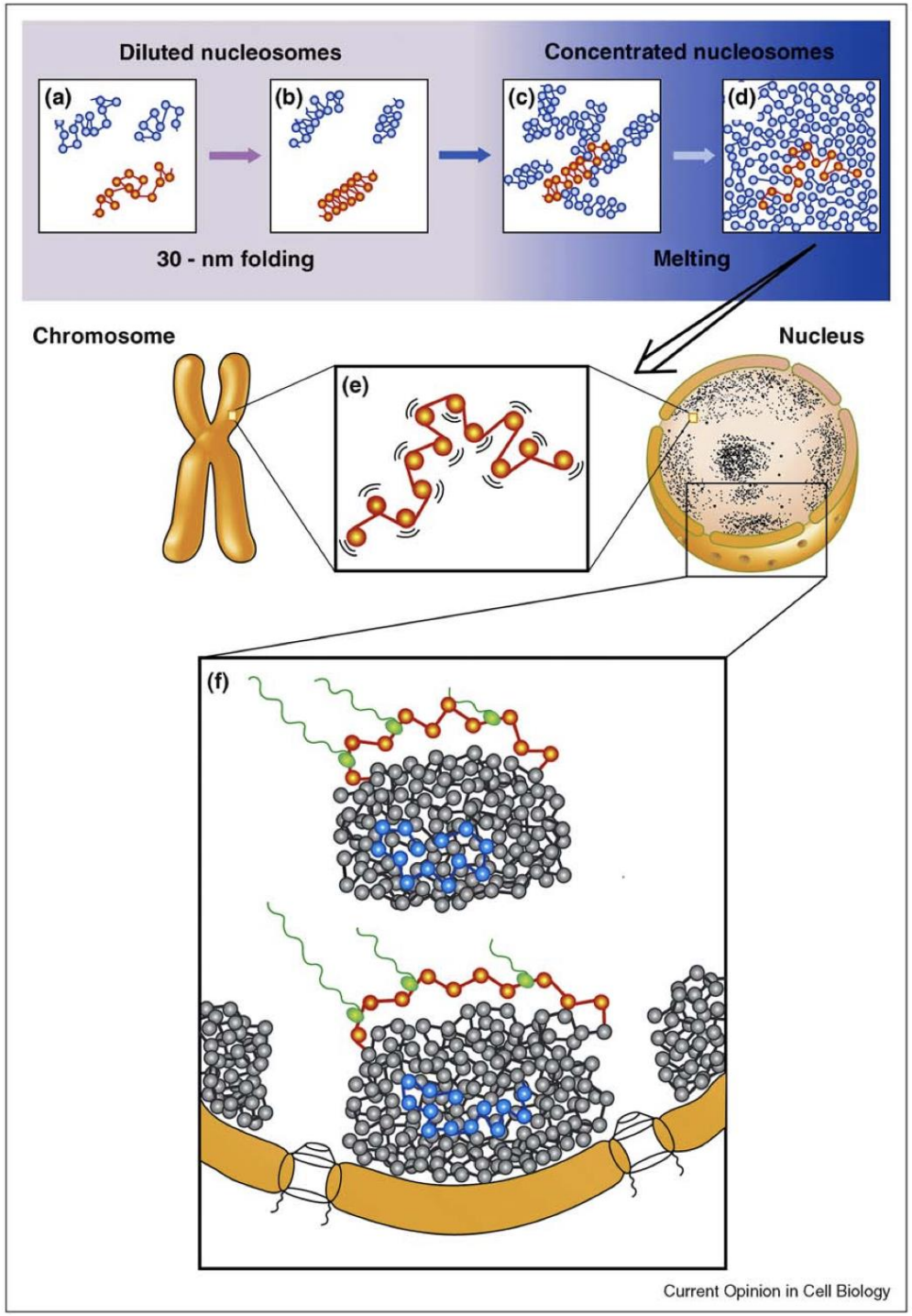
Travers, 2014



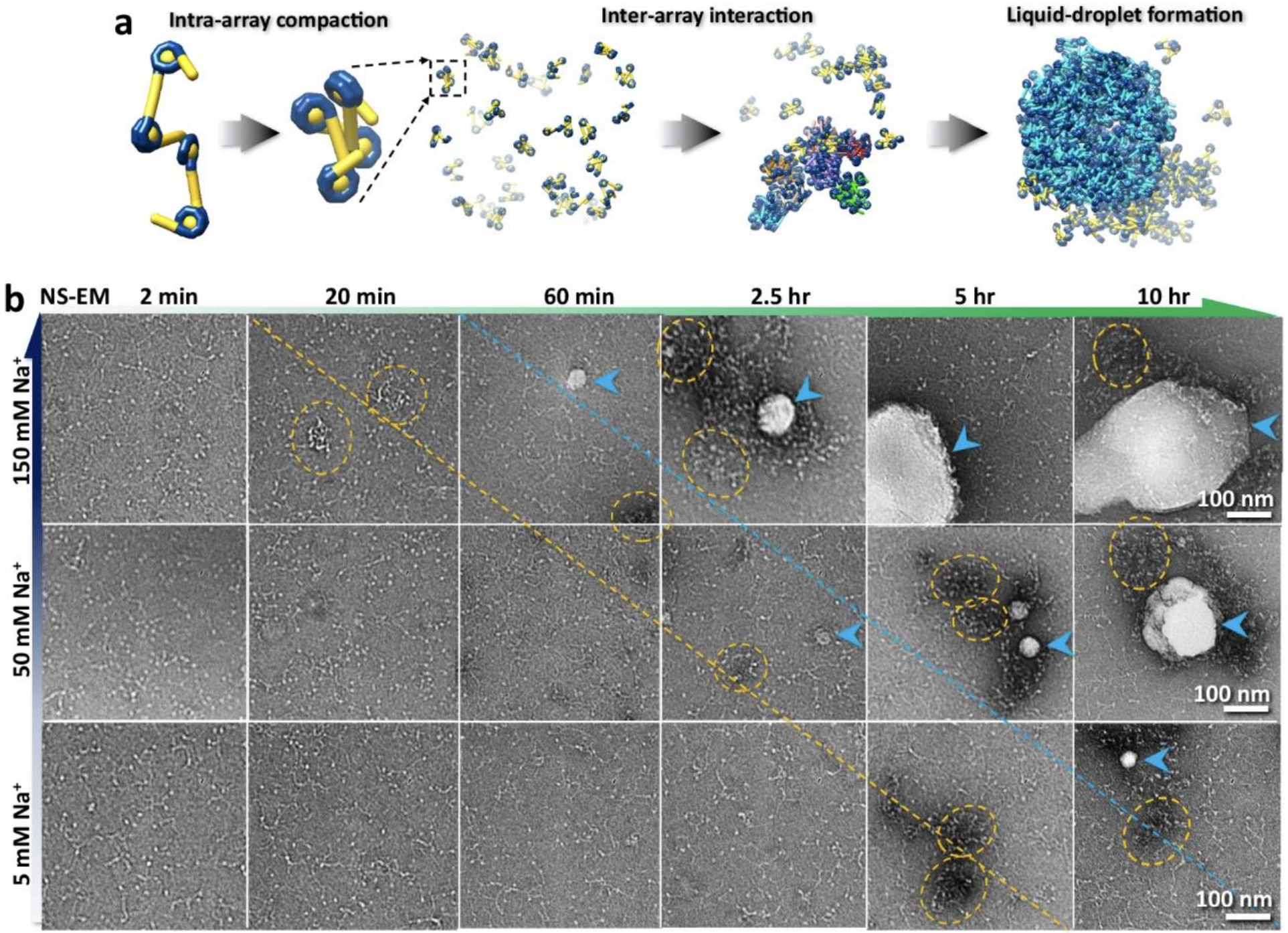


Song et al., 2014

A fibra de 30 nm



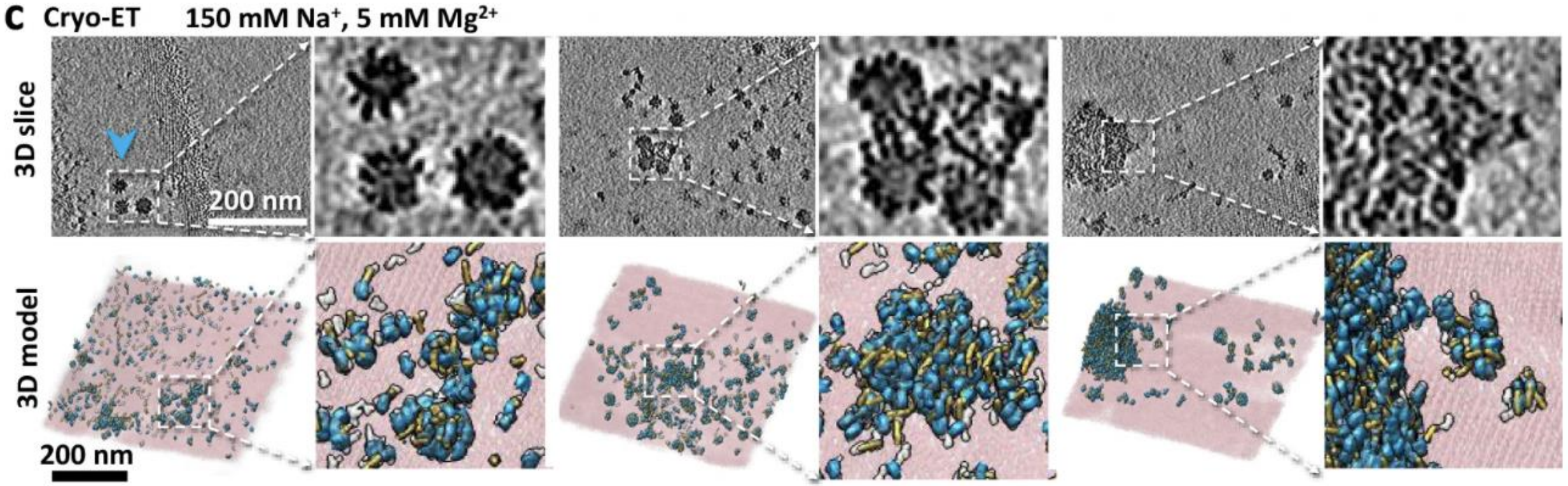
Maeshima, Hihara and Eltsov, 2010

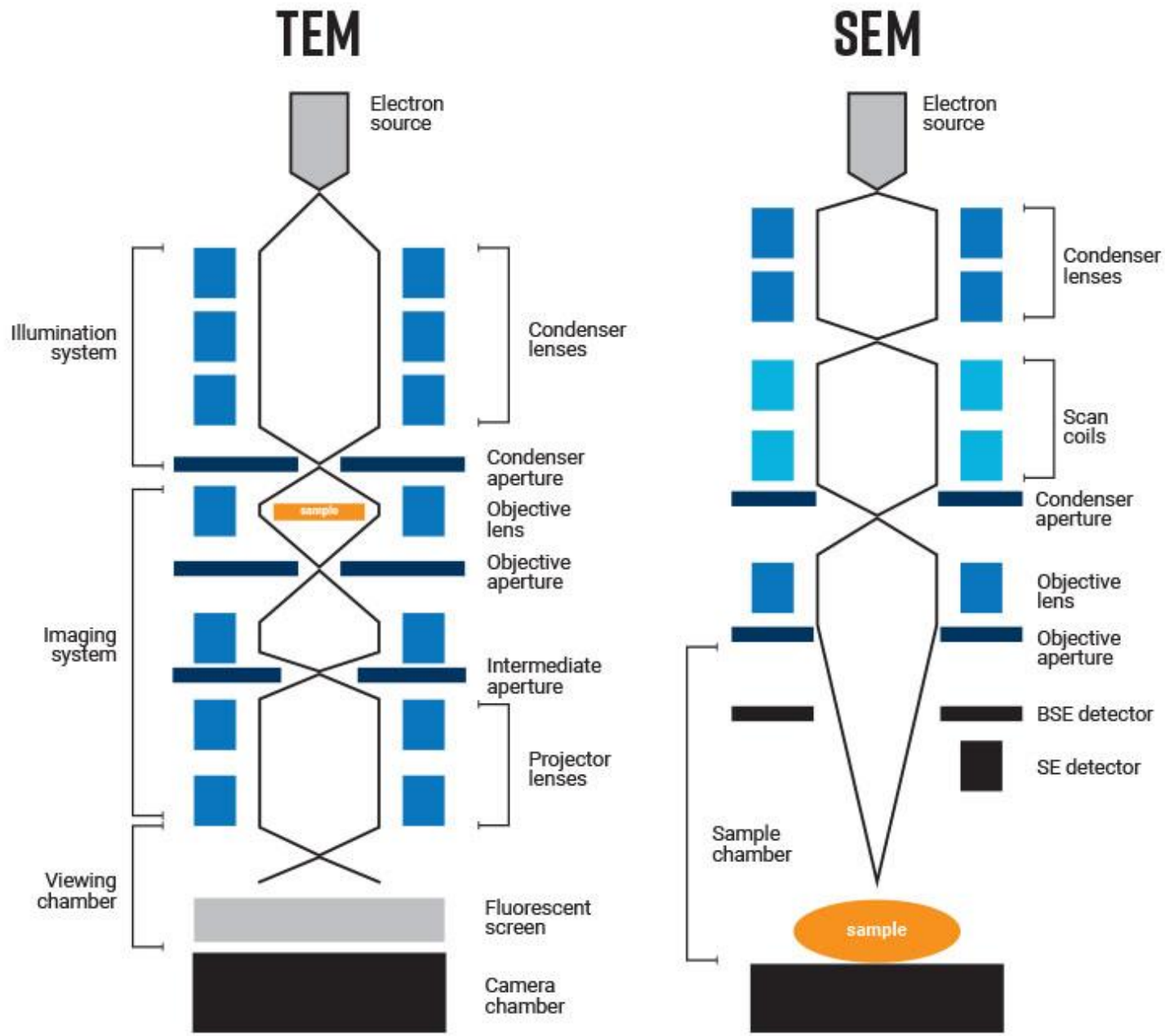


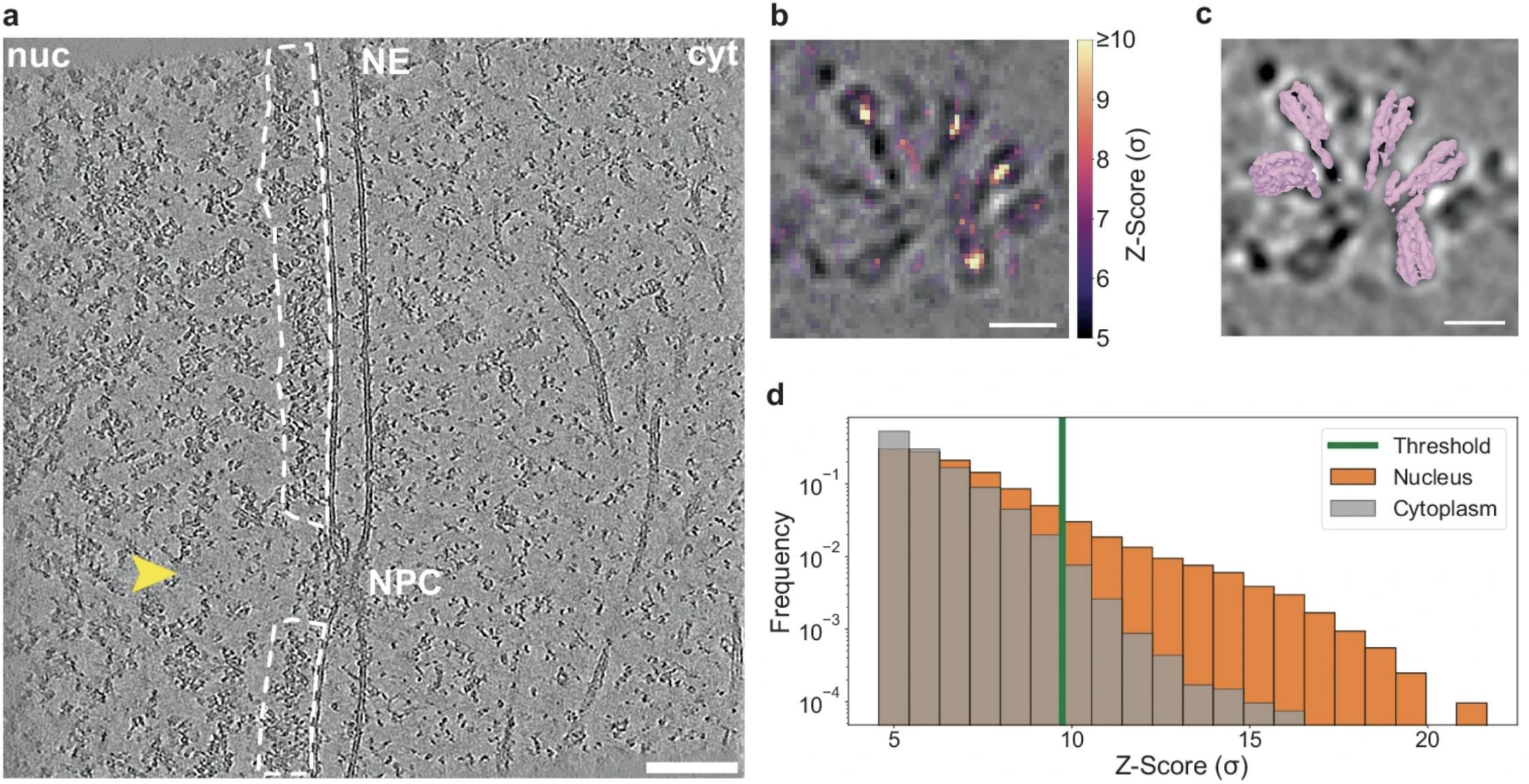
Negative stain electron microscopy

Zhang et al., 2024

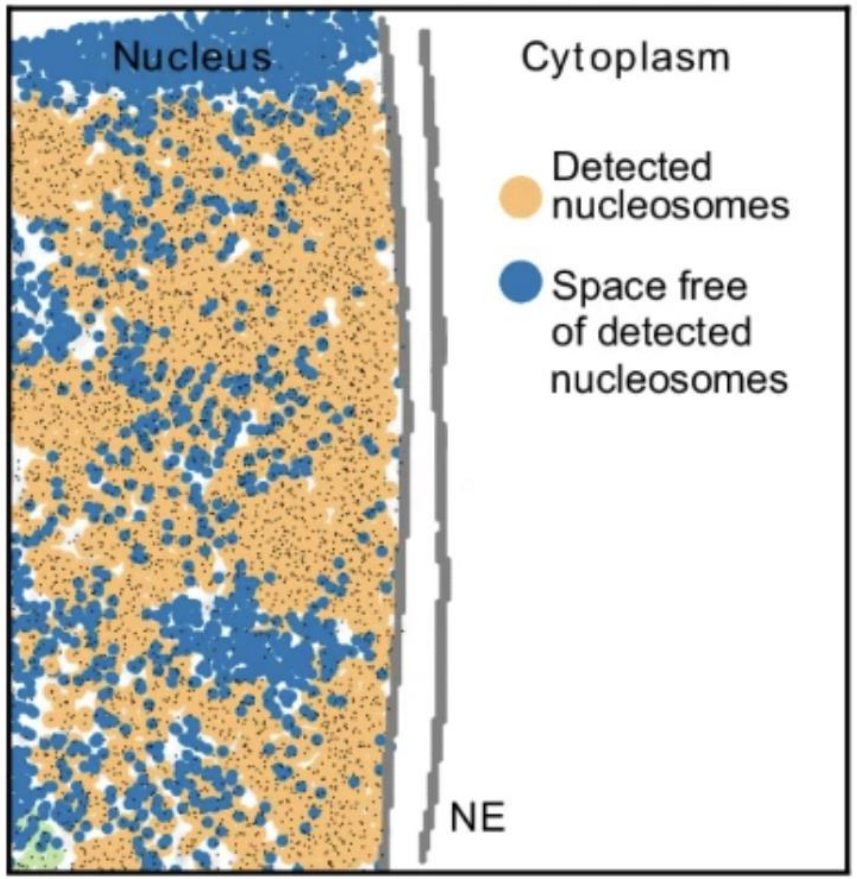
Cryo-electron tomography

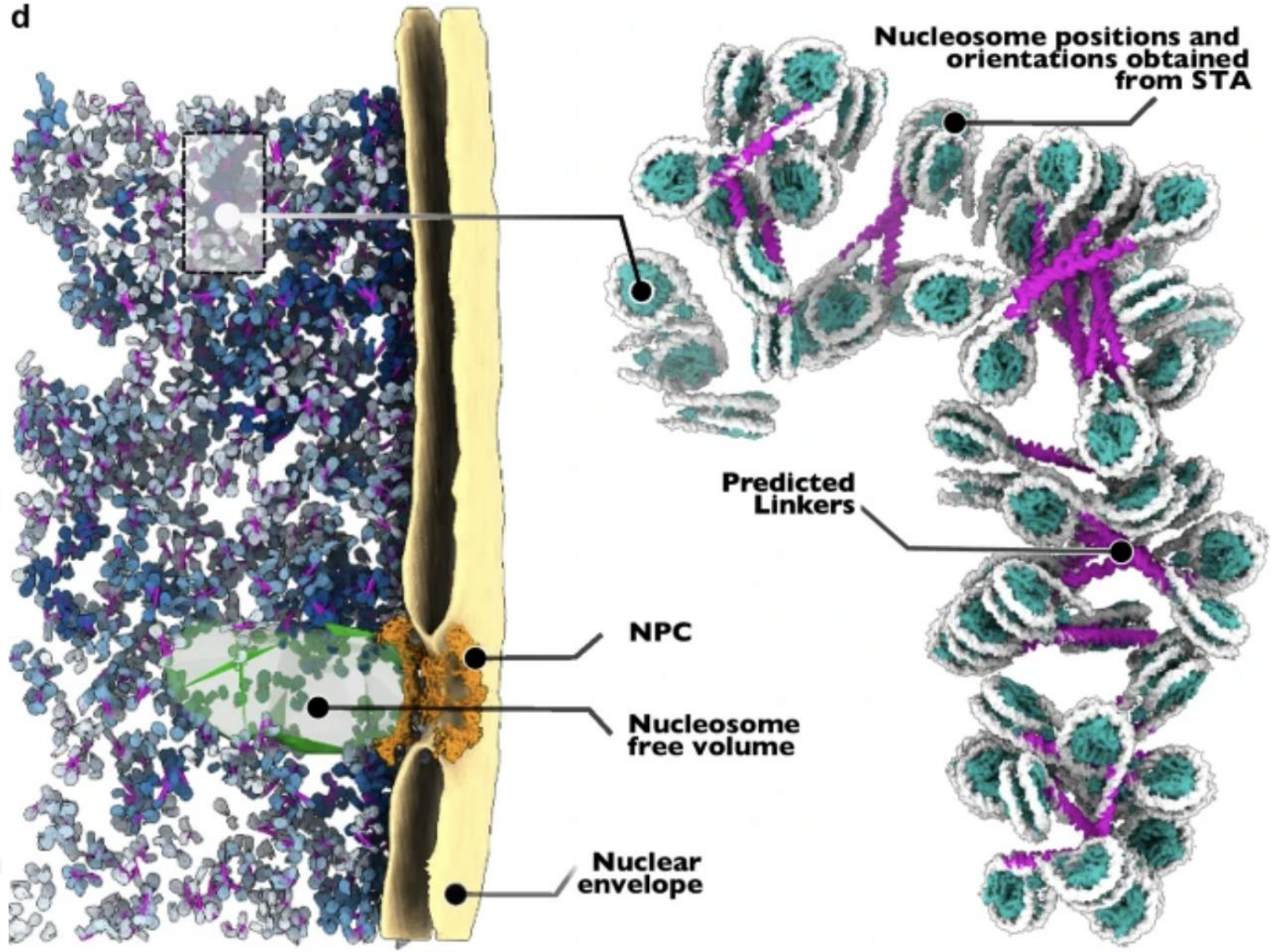



TEM
Electron source
Illumination system
Condenser lenses
Condenser aperture
sample
Objective lens
Objective aperture
Imaging system
Intermediate aperture
Projector lenses
Viewing chamber
Fluorescent screen
Camera chamber
SEM
Electron source
Condenser lenses
Scan coils
Condenser aperture
Objective lens
Objective aperture
BSE detector
SE detector
Sample chamber
sample

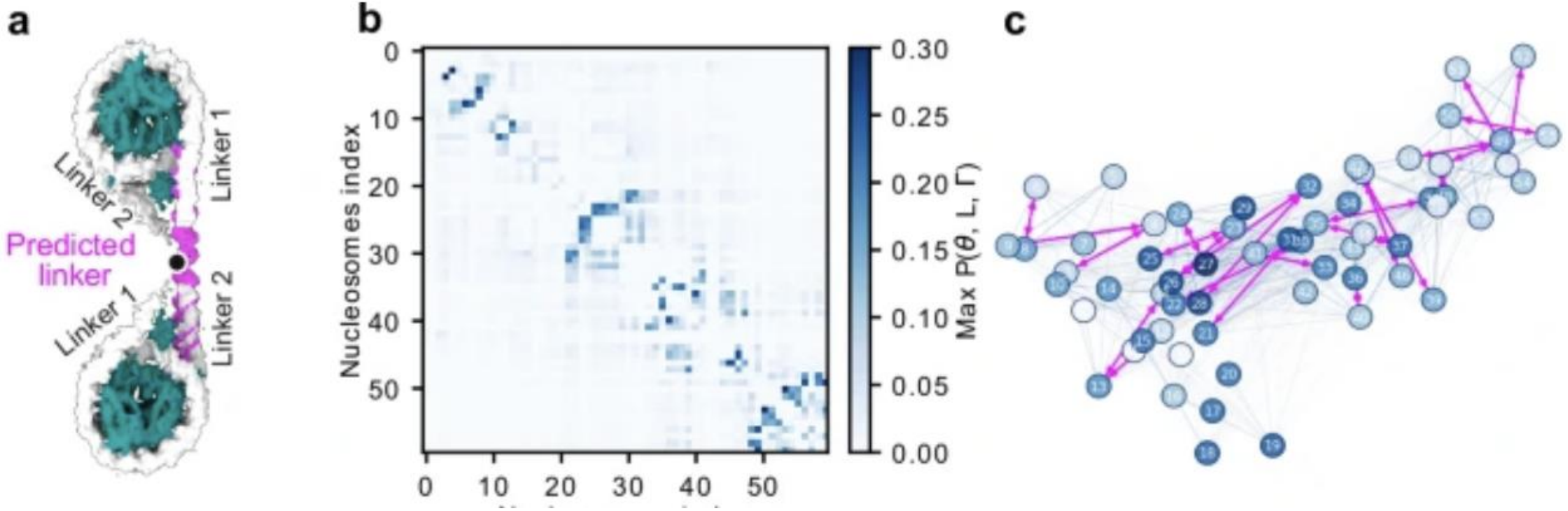


Kreysing et al., 2026


Nucleus
Cytoplasm
Detected nucleosomes
Space free of detected nucleosomes
NE


d
Nucleosome positions and orientations obtained from STA
Predicted Linkers
NPC
Nucleosome free volume
Nuclear envelope

Mapa de contatos



TAGs de histonas

*Feito com IA

Histone mark	Plants	Animals	Main difference
H3K4me3	Active promoters and transcription start sites	Active promoters and transcription start sites	Highly conserved
H3K4me2	Active or permissive promoters and gene-associated regions	Promoters and regulatory regions	Broadly conserved
H3K4me1	Often found in gene bodies and other gene-associated regions; may mark enhancers	Widely used as an enhancer-associated mark	Enhancer interpretation differs
H3K27ac	Associated with active genes and regulatory regions	Strong marker of active promoters and enhancers	Similar function, but enhancer landscapes differ
H3K9ac	Associated with active genes and accessible chrom atin	Associated with active promoters and accessible chrom atin	Highly conserved
H3K36me3	Enriched in transcribed gene bodies	Enriched in transcribed gene bodies	Highly conserved
H3K27me3	Polycomb-mediated repression, particularly of developmental genes	Polycomb-mediated repression, including developmental genes	Highly conserved
H3K9me2	Important for transposable-element silencing and heterochromatin	Associated with repressed chromatin and some heteroc hromatic regions	Particularly prominent in plant TE silencing
H3K9me3	Less prominent in Arabidopsis; distribution varies ac ross plant lineages	Major marker of constitutive heterochromatin in many a nimals	Strong difference in Arabidopsis
H3K27me1	Important in heterochromatin, including many repeti tive regions	Not generally used as the principal constitutive-heterochromatin marker	Particularly notable in plants
H4K20me3	Role and distribution vary; less established as a gen eral plant heterochromatin marker	Associated with compact chromatin and repetitive regio ns in many animals	Evidence and importance differ
H2AK119ub	Involved in Polycomb-associated repression	Involved in Polycomb-associated repression	Broadly conserved, with system-specific details
H2Bub1 (e.g. H2BK120u b in mammals)	Associated with transcription and other chromatin p rocesses	Associated with transcription elongation and histone cro sstalk	Conserved pathway; residue numbering can diff er

Prinpais TAGs

*Feito com IA

Plants: H3K9me2 + H3K27me1

These are particularly important for silencing transposable elements and maintaining constitutive heterochromatin in *Arabidopsis*. doi Annual Revi +1

Animals: H3K9me3

A prominent marker of constitutive heterochromatin in many animals, including mammals.

PMC +1

Enhancers: H3K4me1 and H3K27ac

These are well-established enhancer-associated marks in mammals, but they are less reliable as universal enhancer markers in plants. For example, H3K27ac is associated with active enhancers in rice and maize but is not a reliable active-enhancer hallmark in *Arabidopsis*.

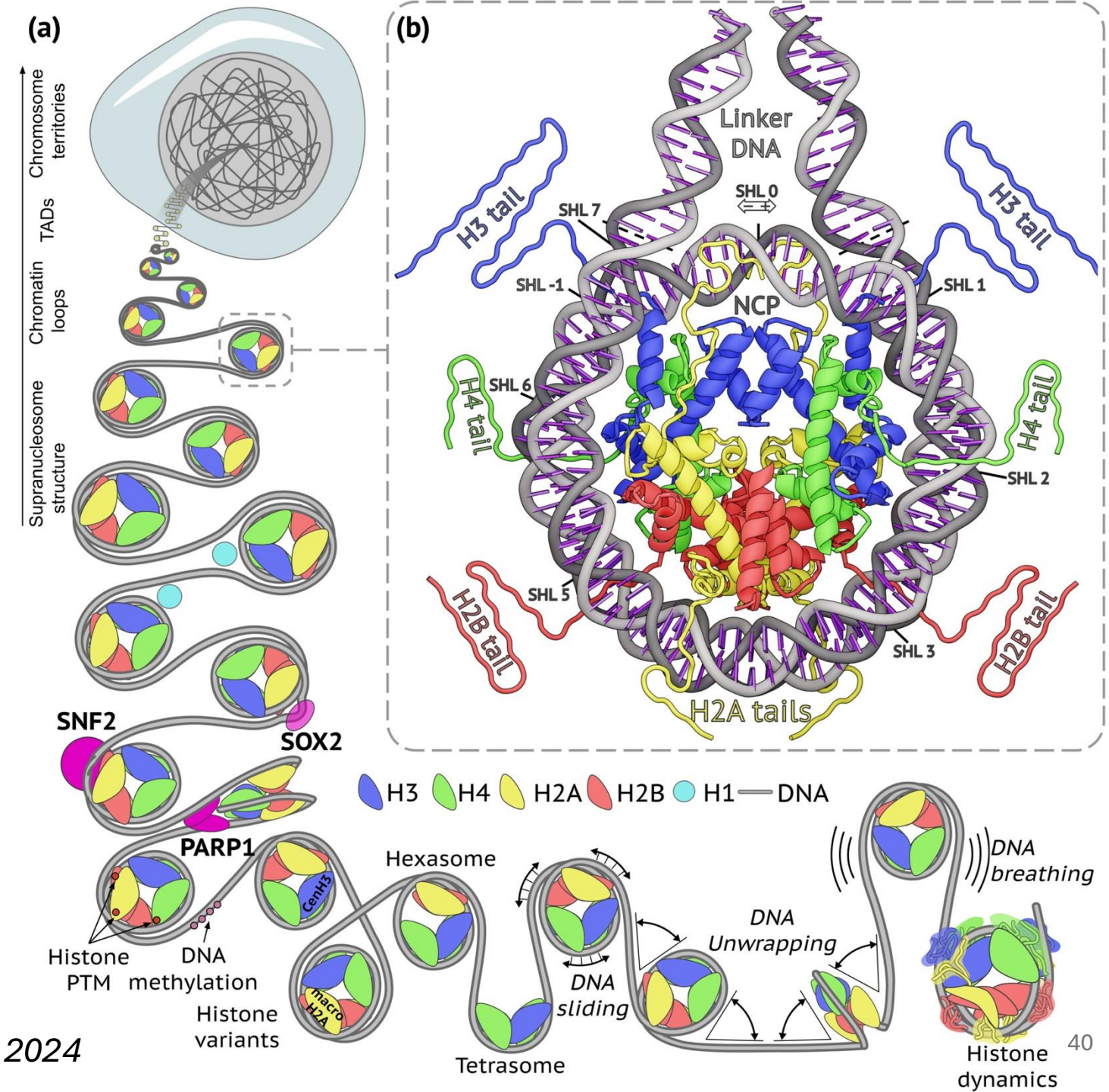
PMC +2

Metilação de DNA

*Feito com IA

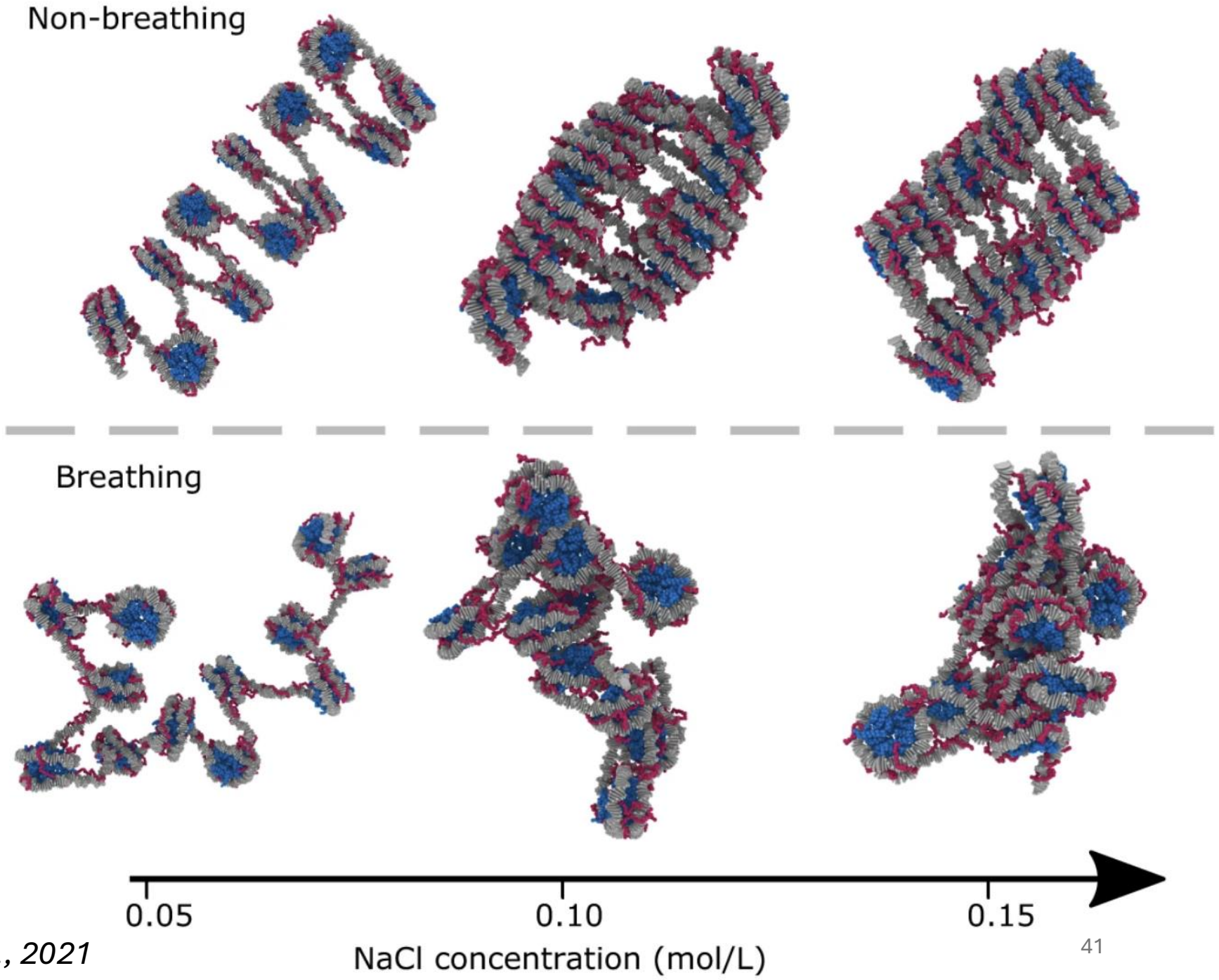
Chromatin state	DNA methylation pattern in plants	DNA methylation pattern in animals (especially mammals)
Active promoter	Usually low DNA methylation near the transcription start site (TSS)	Usually low methylation at active promoters, especially CpG-island promoters
Inactive / repressed promoter	Promoter methylation can be associated with gene repression	Promoter CpG methylation is often associated with stable gene repression
Actively transcribed gene body	CG gene-body methylation is common in some moderately expressed genes	Gene-body CG methylation is common in expressed genes
Constitutive heterochromatin	Often enriched in CG and CHG methylation, with CHH methylation also important in many repeat-rich regions	Often enriched in CG methylation
Transposable-element-rich regions	CG, CHG and CHH methylation can contribute to silencing	CG methylation is a major mechanism of transposable-element repression
Polycomb-repressed chromatin	Frequently associated with H3K27me3; DNA methylation is variable and can be limited at some targets	Frequently associated with H3K27me3; DNA methylation may reinforce repression at some loci
Genomic imprinting	Parent-of-origin-specific methylation occurs in particular regions, notably in endosperm	Parent-of-origin-specific methylation is important at imprinted loci
Broadly active chromatin / euchromatin	Generally lower methylation in regulatory regions, but gene-body methylation can be present	Generally lower methylation at regulatory regions, but gene-body methylation can be present

A dinâmica dos nucleossomos



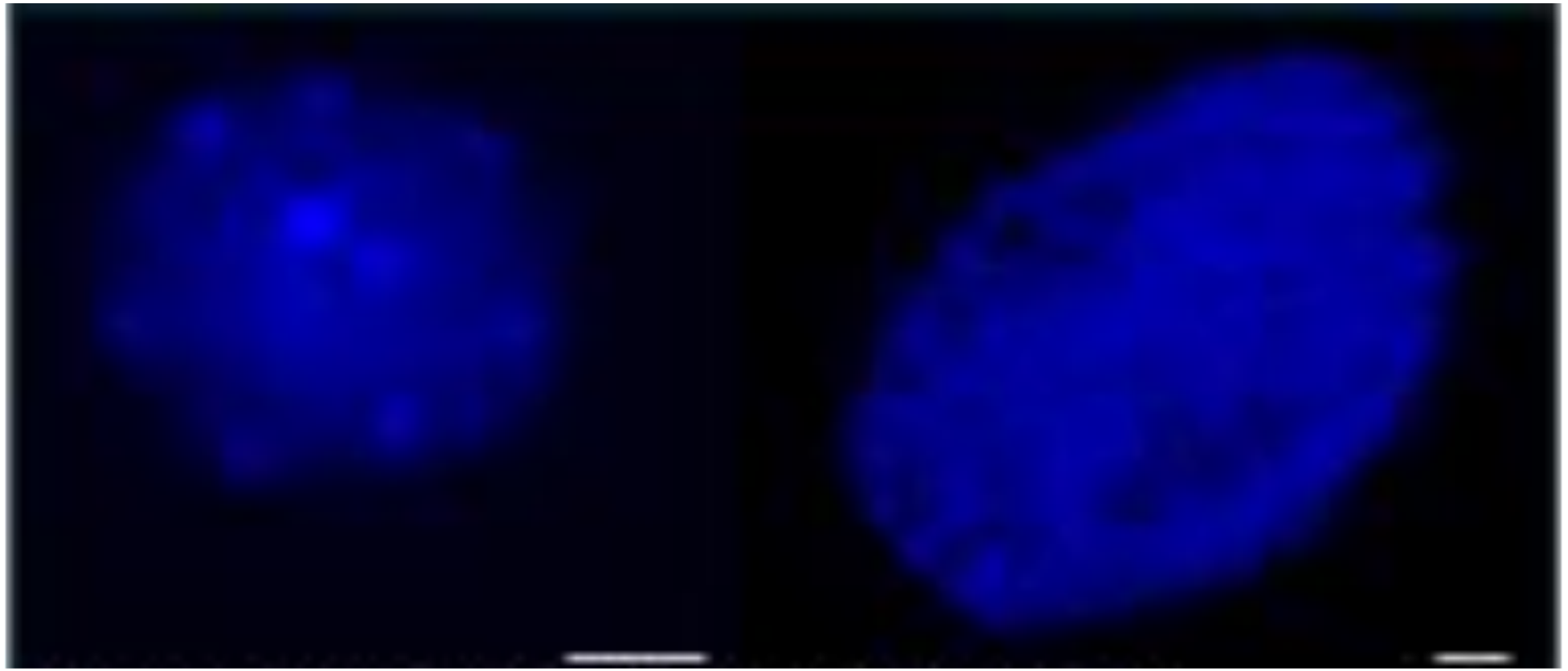
Fedulova et al., 2024

Respiração dos nucleossomos



Farr et al., 2021

Heterocromatina e eu cromatina vistas com DAPI



Arabidopsis

Cevada

Inna Lermontova

Células de folha